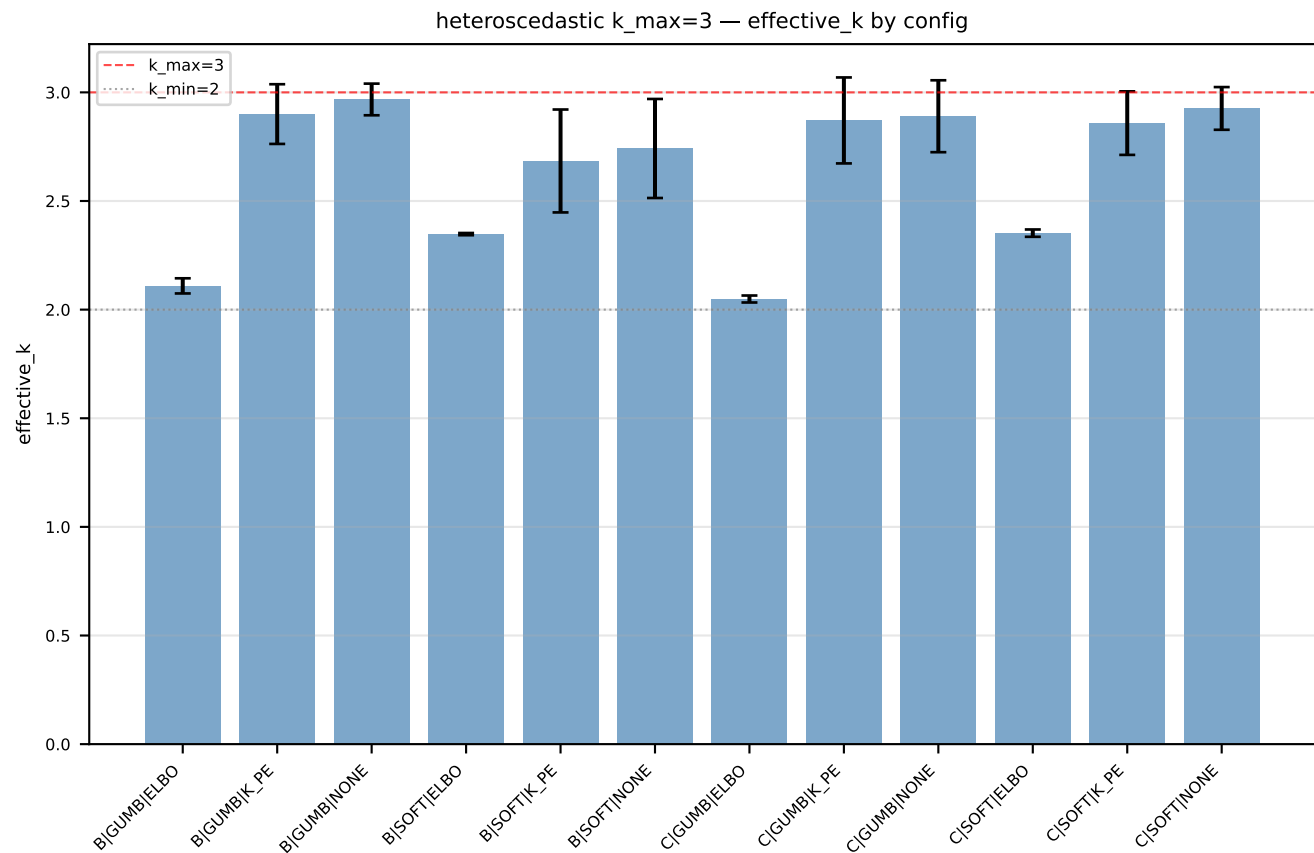
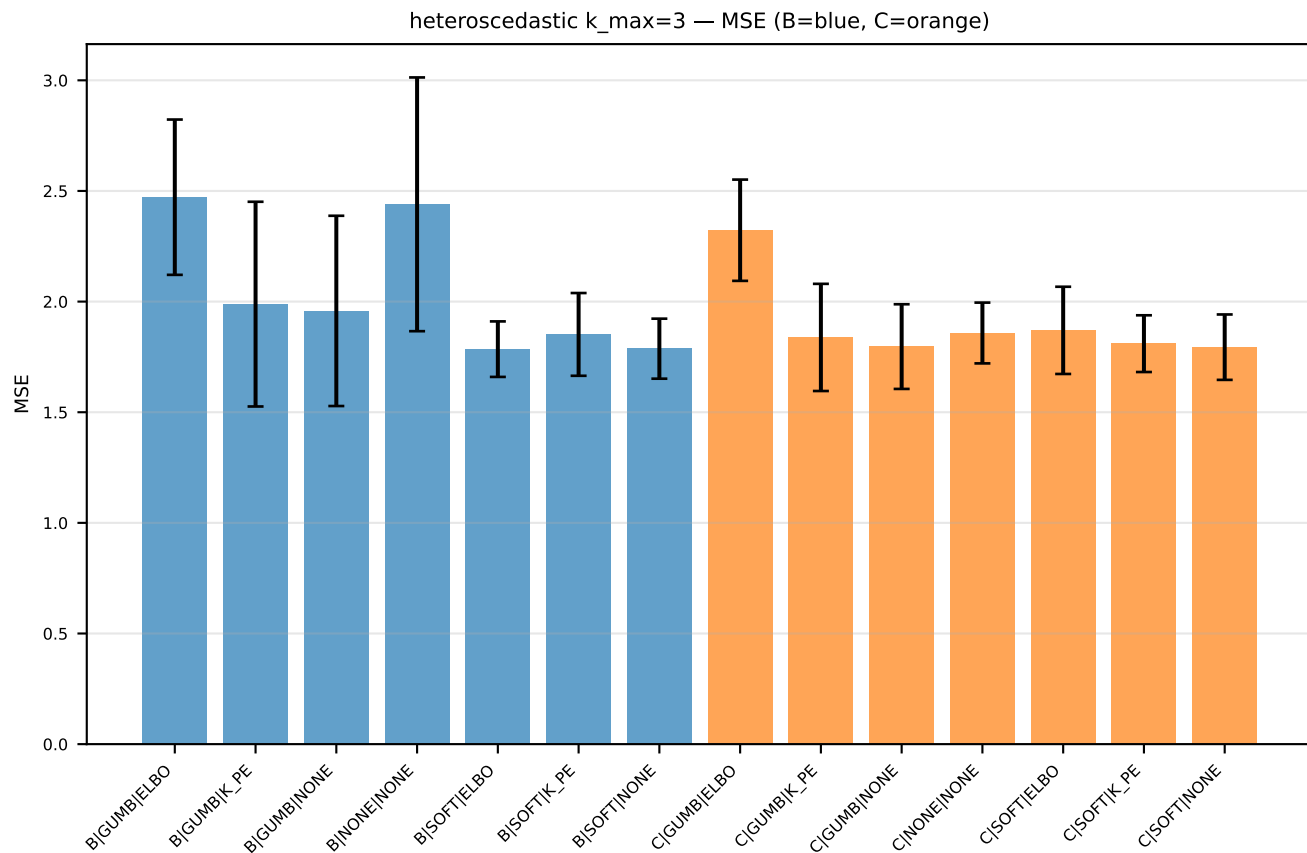
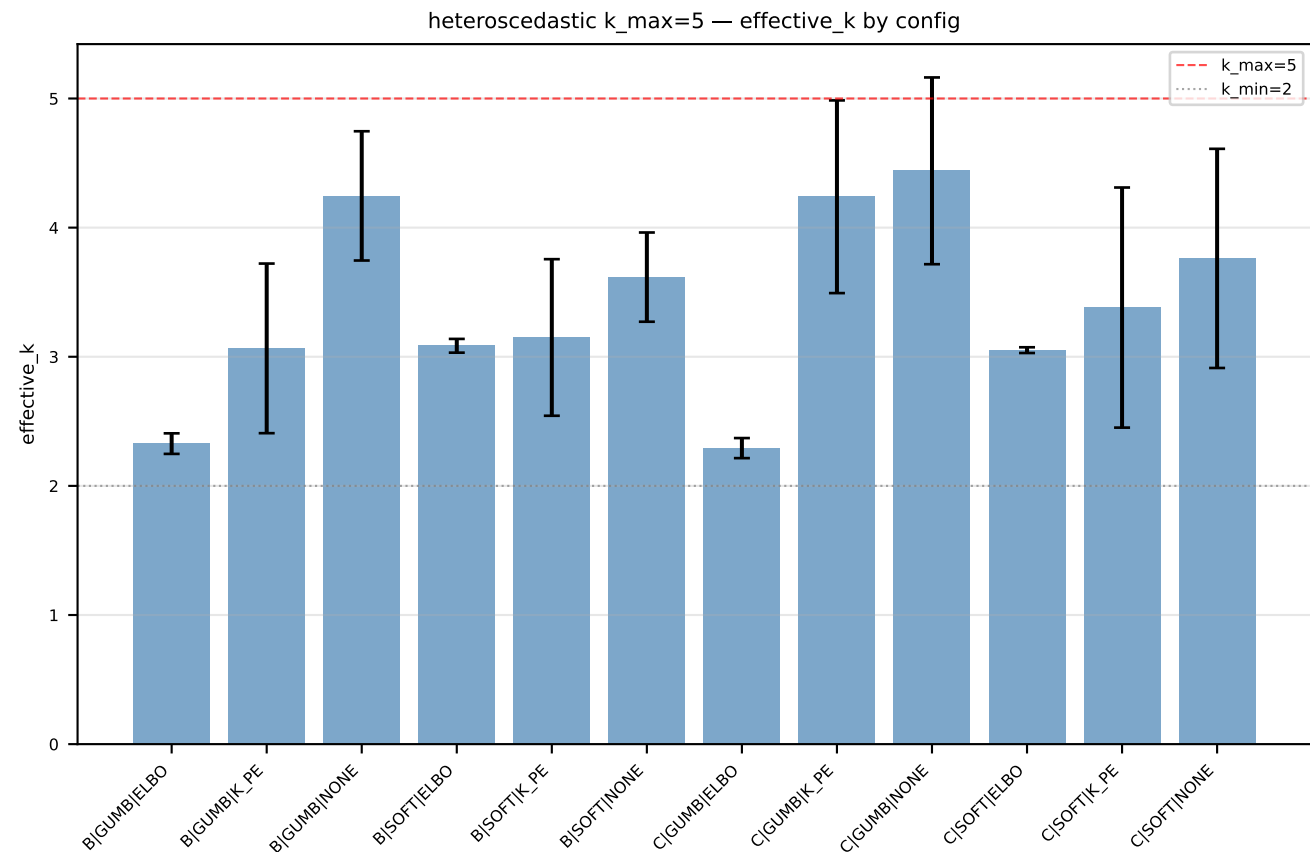
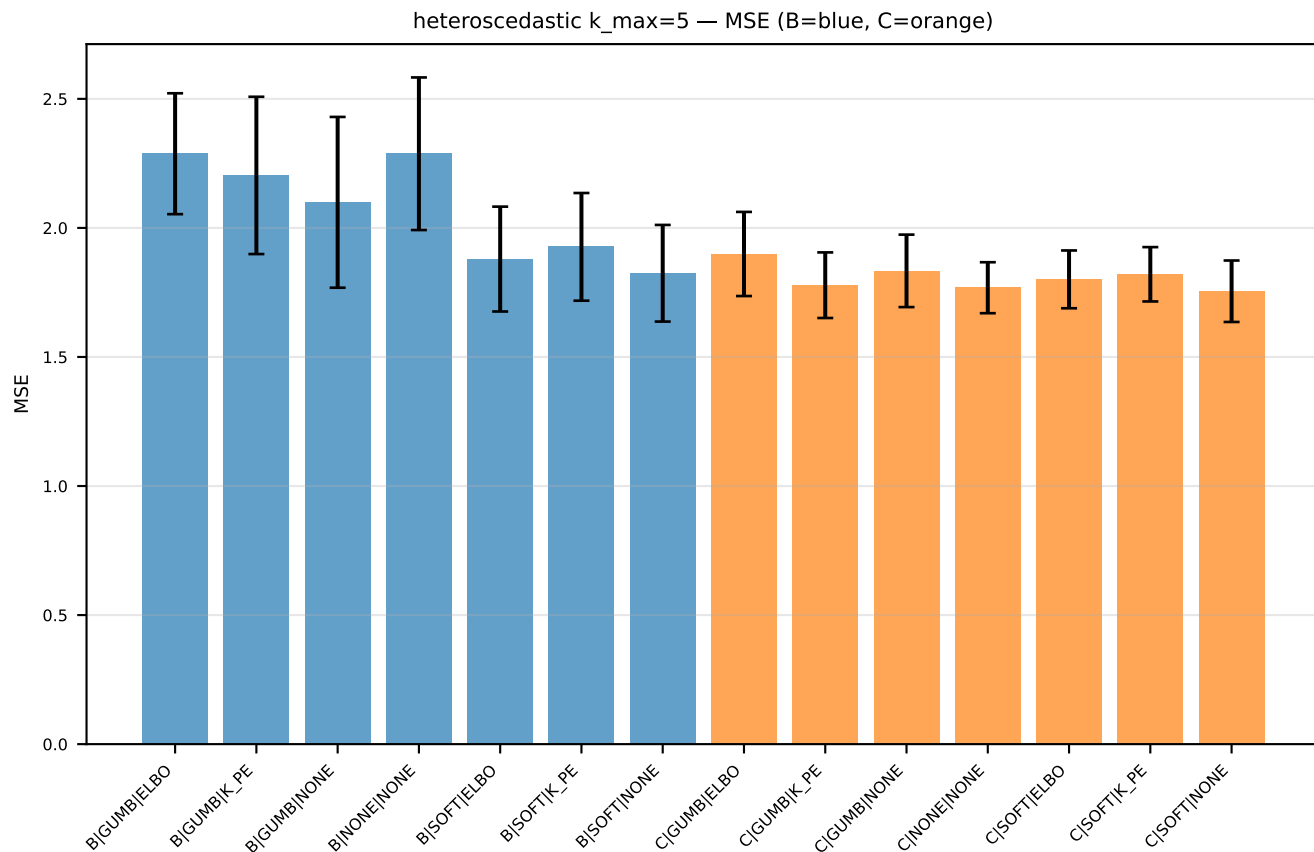


| k_max | cell | dynamic        | k_reg     | mse_mean           | mse_std             | nll_gaussian_mean  | effective_k_mean   | max_p_mid_mean      |
|-------|------|----------------|-----------|--------------------|---------------------|--------------------|--------------------|---------------------|
| 3     | B    | GUMBEL_SOFTMAX | ELBO      | 2.4717957999999998 | 0.35093696885153003 | 1.8237816000000002 | 2.10934            | 0.11884000000000002 |
| 3     | B    | GUMBEL_SOFTMAX | K_PENALTY | 1.9885422          | 0.46258891239382294 | 1.4671144          | 2.90008            | 0.24242             |
| 3     | B    | GUMBEL_SOFTMAX | NONE      | 1.9579307999999997 | 0.42991708769796527 | 1.4564506          | 2.9674             | 0.23364             |
| 3     | B    | NONE           | NONE      | 2.4396215999999997 | 0.5733738243635649  | 1.7036368          | 3.0                | 0.09106             |
| 3     | B    | SOFT_GATING    | ELBO      | 1.7850378          | 0.12544687687304135 | 1.4260394          | 2.3476             | 0.281               |
| 3     | B    | SOFT_GATING    | K_PENALTY | 1.8517038          | 0.1870056660577428  | 1.435503           | 2.68426            | 0.19290000000000002 |
| 3     | B    | SOFT_GATING    | NONE      | 1.7871768          | 0.13560584997041986 | 1.4049718          | 2.7417000000000002 | 0.19324             |
| 3     | C    | GUMBEL_SOFTMAX | ELBO      | 2.3225004          | 0.22883623712668422 | 1.7817042          | 2.0488799999999996 | 0.15064             |
| 3     | C    | GUMBEL_SOFTMAX | K_PENALTY | 1.8380498000000003 | 0.24215045899956503 | 1.475809           | 2.8708400000000003 | 0.34054             |
| 3     | C    | GUMBEL_SOFTMAX | NONE      | 1.7966438          | 0.1913334393910275  | 1.4582252          | 2.8901399999999997 | 0.34016             |
| 3     | C    | NONE           | NONE      | 1.858069           | 0.1373295004032273  | 1.6668998          | 3.0                | 0.31508             |
| 3     | C    | SOFT_GATING    | ELBO      | 1.8699966          | 0.19708909955347603 | 1.4456788          | 2.3520600000000003 | 0.27794             |
| 3     | C    | SOFT_GATING    | K_PENALTY | 1.8099078          | 0.12820882051442473 | 1.4653032          | 2.85846            | 0.35548             |
| 3     | C    | SOFT_GATING    | NONE      | 1.7940108          | 0.1478089848273779  | 1.4630836          | 2.92608            | 0.33972             |
| 5     | B    | GUMBEL_SOFTMAX | ELBO      | 2.2874576          | 0.2341922528560669  | 1.7731222000000002 | 2.3273800000000002 | 0.5676              |
| 5     | B    | GUMBEL_SOFTMAX | K_PENALTY | 2.2033859999999996 | 0.3046891777738421  | 1.550174           | 3.06494            | 0.58714             |
| 5     | B    | GUMBEL_SOFTMAX | NONE      | 2.0991646          | 0.33083156173618017 | 1.5411540000000001 | 4.24586            | 0.50884             |
| 5     | B    | NONE           | NONE      | 2.2873706          | 0.29560866483714565 | 1.6847676          | 5.0                | 0.56272             |
| 5     | B    | SOFT_GATING    | ELBO      | 1.8793042          | 0.20312277791449188 | 1.442985           | 3.0849800000000003 | 0.58108             |
| 5     | B    | SOFT_GATING    | K_PENALTY | 1.9266524          | 0.20862076716688582 | 1.4529188          | 3.1494             | 0.52254             |
| 5     | B    | SOFT_GATING    | NONE      | 1.824348           | 0.1872077742402809  | 1.4796386          | 3.61678            | 0.5631              |
| 5     | C    | GUMBEL_SOFTMAX | ELBO      | 1.8990088          | 0.1629424292800989  | 1.6704830000000002 | 2.29258            | 0.2187199999999997  |
| 5     | C    | GUMBEL_SOFTMAX | K_PENALTY | 1.7780660000000001 | 0.12707818535256155 | 1.4601668          | 4.2391             | 0.38308             |
| 5     | C    | GUMBEL_SOFTMAX | NONE      | 1.8336078          | 0.14039554639553198 | 1.4983236          | 4.43992            | 0.34926             |
| 5     | C    | NONE           | NONE      | 1.7683091999999998 | 0.09874273374633698 | 1.6008422          | 5.0                | 0.37454             |
| 5     | C    | SOFT_GATING    | ELBO      | 1.8006742          | 0.11196055107581418 | 1.4121578          | 3.05124            | 0.41852             |
| 5     | C    | SOFT_GATING    | K_PENALTY | 1.8203638000000002 | 0.10524034081900344 | 1.475041           | 3.3809799999999997 | 0.40692000000000006 |
| 5     | C    | SOFT_GATING    | NONE      | 1.7546427999999998 | 0.11905701378415305 | 1.4106916          | 3.7616799999999997 | 0.30938             |
| 7     | B    | GUMBEL_SOFTMAX | ELBO      | 2.2240786          | 0.3386631690238843  | 1.7363228          | 2.68418            | 0.19290000000000002 |
| 7     | B    | GUMBEL_SOFTMAX | K_PENALTY | 1.9143245999999998 | 0.1282841275170082  | 1.502674           | 5.13296            | 0.25270000000000004 |
| 7     | B    | GUMBEL_SOFTMAX | NONE      | 1.8799108          | 0.19762703688311473 | 1.4688062          | 6.016360000000001  | 0.2956              |
| 7     | B    | NONE           | NONE      | 2.1354568          | 0.3270713757235569  | 1.5798717999999998 | 7.0                | 0.09462             |
| 7     | B    | SOFT_GATING    | ELBO      | 1.8137568000000002 | 0.14528322192600215 | 1.4417138          | 3.7930599999999997 | 0.45116000000000006 |
| 7     | B    | SOFT_GATING    | K_PENALTY | 1.8762508          | 0.16728572926792054 | 1.4829628          | 4.69032            | 0.27188             |
| 7     | B    | SOFT_GATING    | NONE      | 1.8209578          | 0.13657616981633364 | 1.4779426          | 5.51492            | 0.36384             |
| 7     | C    | GUMBEL_SOFTMAX | ELBO      | 1.8149895999999999 | 0.1662022720792348  | 1.6618526          | 2.57024            | 0.2009599999999997  |
| 7     | C    | GUMBEL_SOFTMAX | K_PENALTY | 1.805677           | 0.1109453294465341  | 1.4396942          | 5.25462            | 0.32504             |
| 7     | C    | GUMBEL_SOFTMAX | NONE      | 1.8409442000000003 | 0.1279122460857443  | 1.4277504          | 5.803459999999999  | 0.28364             |
| 7     | C    | NONE           | NONE      | 1.7277026          | 0.10106313749483539 | 1.5927012          | 7.0                | 0.24508000000000002 |
| 7     | C    | SOFT_GATING    | ELBO      | 1.8059798          | 0.2191040116193676  | 1.4494644          | 3.82742            | 0.09516             |
| 7     | C    | SOFT_GATING    | K_PENALTY | 1.783197           | 0.24726327786187738 | 1.3988158          | 5.08286            | 0.42026             |
| 7     | C    | SOFT_GATING    | NONE      | 1.7604692          | 0.11156899737247802 | 1.4285548000000001 | 5.534879999999999  | 0.3969              |

heteroscedastic k\_max=3

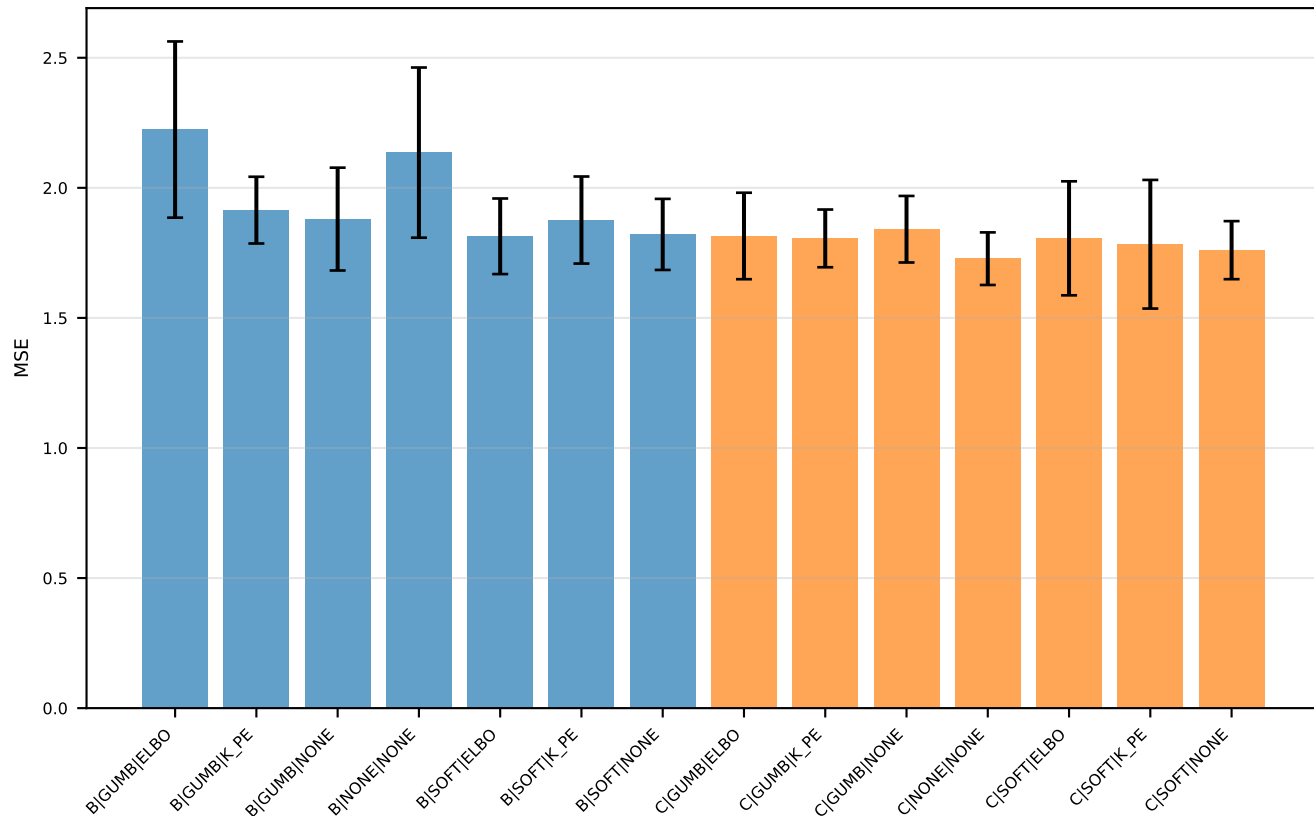


heteroscedastic k\_max=5

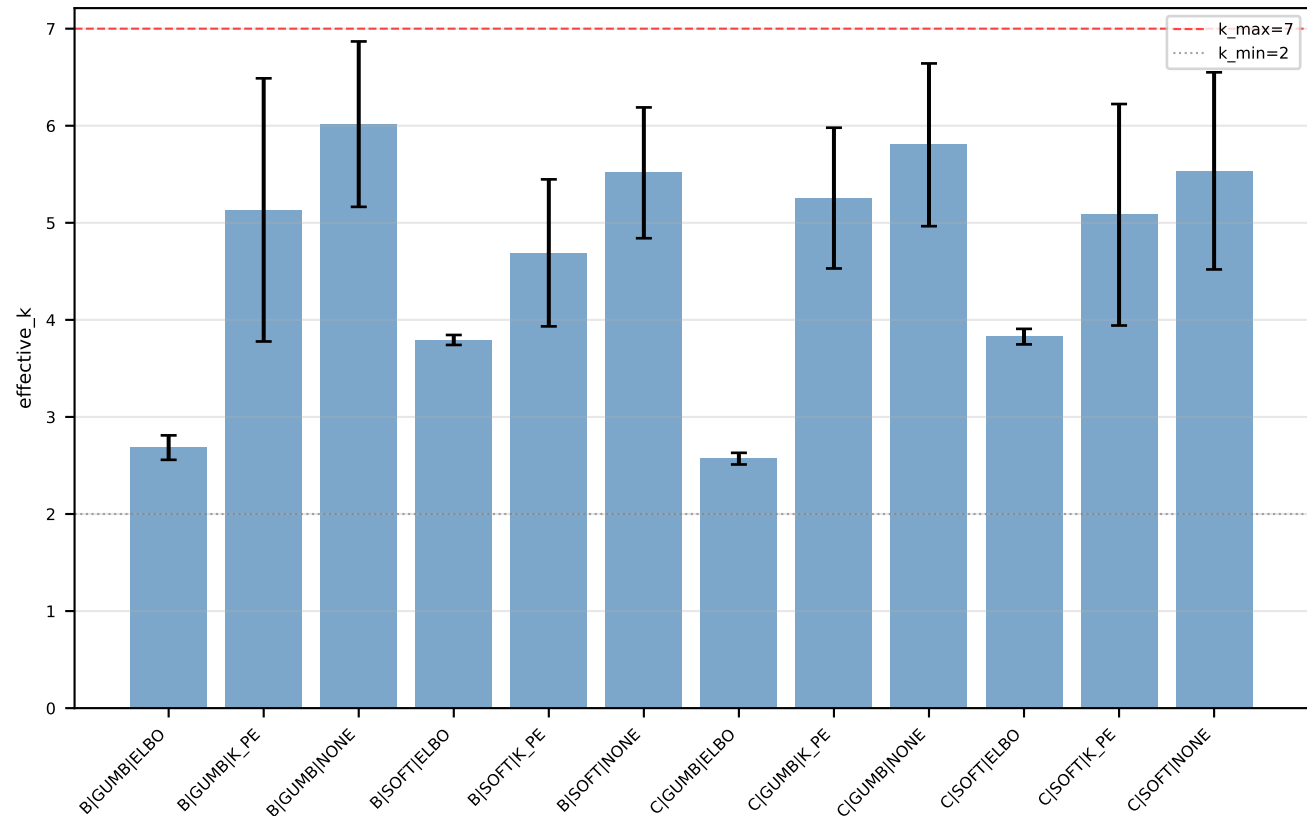


# heteroscedastic k\_max=7

heteroscedastic k\_max=7 — MSE (B=blue, C=orange)



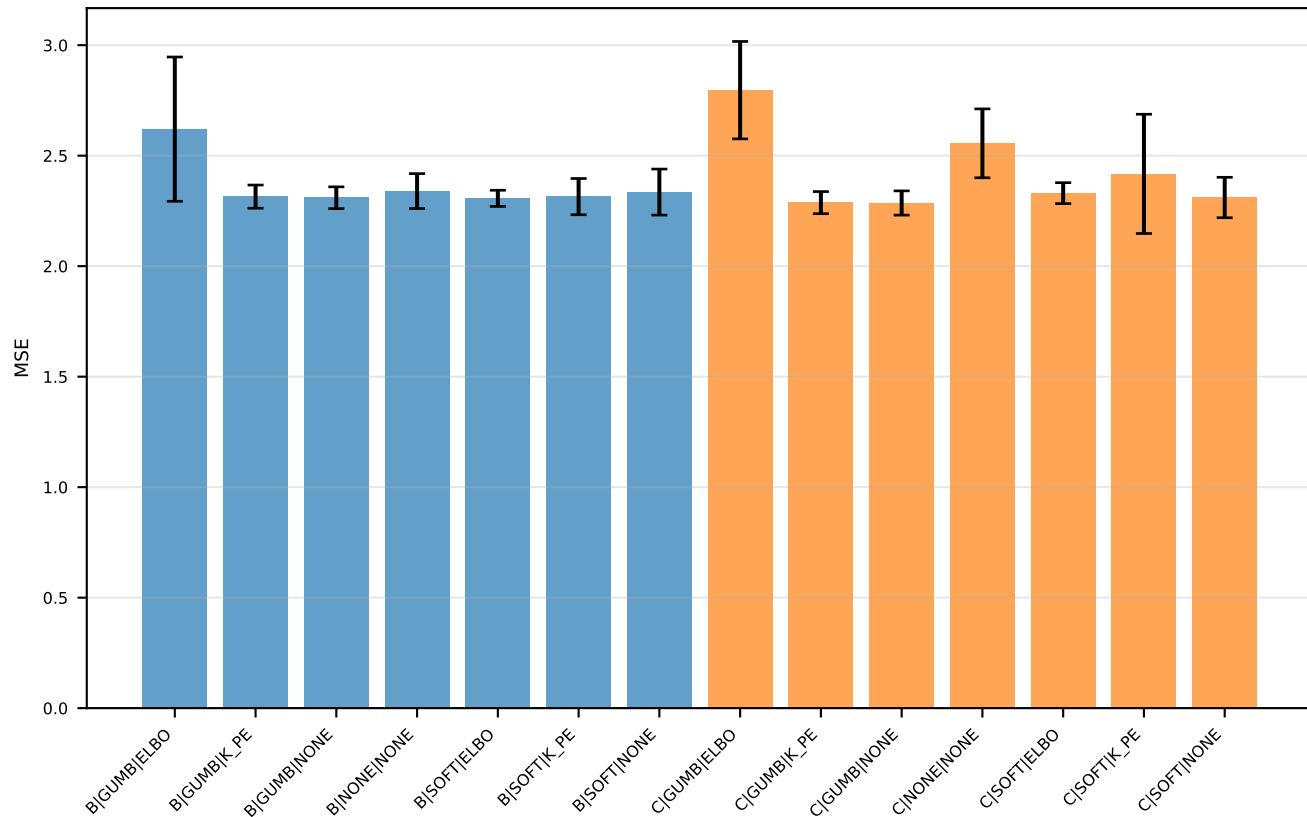
heteroscedastic k\_max=7 — effective\_k by config



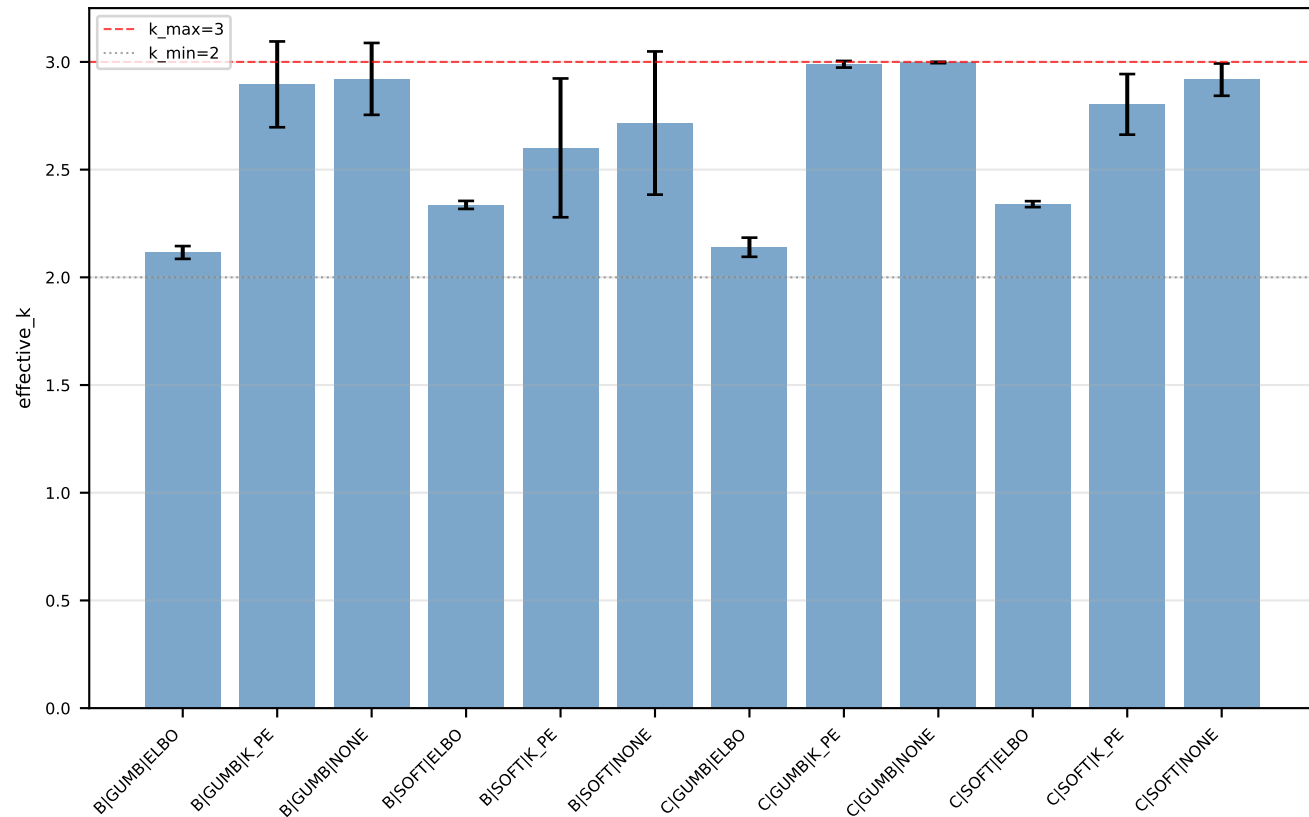
| k_max | cell | dynamic        | k_reg     | mse_mean           | mse_std              | nll_gaussian_mean  | effective_k_mean   | max_p_mid_mean      |
|-------|------|----------------|-----------|--------------------|----------------------|--------------------|--------------------|---------------------|
| 3     | B    | GUMBEL_SOFTMAX | ELBO      | 2.620092           | 0.3264706591839457   | 1.9061990000000002 | 2.11542            | 0.3760199999999997  |
| 3     | B    | GUMBEL_SOFTMAX | K_PENALTY | 2.3147235999999998 | 0.05245539564525279  | 1.8397832          | 2.8959             | 0.42590000000000006 |
| 3     | B    | GUMBEL_SOFTMAX | NONE      | 2.3097164          | 0.04922003241364236  | 1.8386848          | 2.92118            | 0.43034             |
| 3     | B    | NONE           | NONE      | 2.3398524000000003 | 0.07897884563476457  | 1.8469356000000001 | 3.0                | 0.28650000000000003 |
| 3     | B    | SOFT_GATING    | ELBO      | 2.306907           | 0.03663859009296074  | 1.8382638          | 2.33622            | 0.3832              |
| 3     | B    | SOFT_GATING    | K_PENALTY | 2.3147390000000003 | 0.08195327679232842  | 1.8409424          | 2.60106            | 0.33492             |
| 3     | B    | SOFT_GATING    | NONE      | 2.3353279999999996 | 0.10413345037018602  | 1.8454443999999999 | 2.7163             | 0.30404             |
| 3     | C    | GUMBEL_SOFTMAX | ELBO      | 2.796468           | 0.22048042849309799  | 1.9504416          | 2.13976            | 0.27266             |
| 3     | C    | GUMBEL_SOFTMAX | K_PENALTY | 2.2875172          | 0.05003065037254639  | 1.8333994          | 2.9892             | 0.36544             |
| 3     | C    | GUMBEL_SOFTMAX | NONE      | 2.2858718          | 0.05483146737686295  | 1.833135           | 2.9974800000000004 | 0.40953999999999996 |
| 3     | C    | NONE           | NONE      | 2.5558794          | 0.15586271232818946  | 1.9132946          | 3.0                | 0.31706             |
| 3     | C    | SOFT_GATING    | ELBO      | 2.3304169999999997 | 0.04735376045785597  | 1.8435638          | 2.33978            | 0.2334              |
| 3     | C    | SOFT_GATING    | K_PENALTY | 2.4176354          | 0.2698723716505636   | 1.8642632          | 2.80304            | 0.30326             |
| 3     | C    | SOFT_GATING    | NONE      | 2.310705           | 0.09146876890775328  | 1.8404466          | 2.9177799999999996 | 0.33168000000000003 |
| 5     | B    | GUMBEL_SOFTMAX | ELBO      | 2.3541894          | 0.0765768915718049   | 1.8498246000000003 | 2.38958            | 0.51244             |
| 5     | B    | GUMBEL_SOFTMAX | K_PENALTY | 2.3222633999999998 | 0.049544575927138595 | 1.8415362000000002 | 2.86152            | 0.46628             |
| 5     | B    | GUMBEL_SOFTMAX | NONE      | 2.3437031999999998 | 0.059270690886812    | 1.8458288          | 4.29274            | 0.48460000000000003 |
| 5     | B    | NONE           | NONE      | 2.3356316          | 0.05945100164252916  | 1.845183           | 5.0                | 0.42024             |
| 5     | B    | SOFT_GATING    | ELBO      | 2.313685           | 0.051629342659964036 | 1.8389376          | 3.04174            | 0.5217              |
| 5     | B    | SOFT_GATING    | K_PENALTY | 2.3189152          | 0.053626277632332665 | 1.840811           | 2.5077399999999996 | 0.49328000000000005 |
| 5     | B    | SOFT_GATING    | NONE      | 2.3383836000000002 | 0.0569038858453445   | 1.8448996000000002 | 3.9932600000000003 | 0.45519999999999994 |
| 5     | C    | GUMBEL_SOFTMAX | ELBO      | 2.5684418          | 0.2628452100252162   | 1.8966176000000001 | 2.26506            | 0.26668             |
| 5     | C    | GUMBEL_SOFTMAX | K_PENALTY | 2.44208            | 0.11294585864475058  | 1.8650238000000001 | 3.6007800000000003 | 0.35607999999999995 |
| 5     | C    | GUMBEL_SOFTMAX | NONE      | 2.3726372          | 0.08996658222751373  | 1.8519036          | 4.24794            | 0.40526             |
| 5     | C    | NONE           | NONE      | 2.5854424000000003 | 0.10420925104711201  | 1.9114385999999999 | 5.0                | 0.45743999999999996 |
| 5     | C    | SOFT_GATING    | ELBO      | 2.3624712000000003 | 0.04970524584689228  | 1.8508928          | 3.04402            | 0.3076              |
| 5     | C    | SOFT_GATING    | K_PENALTY | 2.352643           | 0.055619009623509054 | 1.8481328000000001 | 3.4962400000000002 | 0.30446             |
| 5     | C    | SOFT_GATING    | NONE      | 2.3183508          | 0.05983158196721845  | 1.8400732000000002 | 4.297739999999999  | 0.40406             |
| 7     | B    | GUMBEL_SOFTMAX | ELBO      | 2.3956817999999998 | 0.108716519104504    | 1.8612392          | 2.6139400000000004 | 0.21526             |
| 7     | B    | GUMBEL_SOFTMAX | K_PENALTY | 2.3161506          | 0.06738472128605999  | 1.8407938000000001 | 3.5142399999999996 | 0.28468             |
| 7     | B    | GUMBEL_SOFTMAX | NONE      | 2.3492812          | 0.05996702779611483  | 1.8469924          | 6.10986            | 0.34504             |
| 7     | B    | NONE           | NONE      | 2.3190356000000003 | 0.05727818671885476  | 1.8408324          | 7.0                | 0.13277999999999998 |
| 7     | B    | SOFT_GATING    | ELBO      | 2.2818424          | 0.04220838273731886  | 1.8326328          | 3.7719             | 0.40278             |
| 7     | B    | SOFT_GATING    | K_PENALTY | 2.321592           | 0.08401900649555422  | 1.841196           | 2.58452            | 0.50852             |
| 7     | B    | SOFT_GATING    | NONE      | 2.2891188          | 0.04984210422624612  | 1.8339576000000002 | 5.80096            | 0.32474             |
| 7     | C    | GUMBEL_SOFTMAX | ELBO      | 2.5618572          | 0.08376567806506424  | 1.8928764          | 2.53898            | 0.1614              |
| 7     | C    | GUMBEL_SOFTMAX | K_PENALTY | 2.3462492          | 0.10393747852771877  | 1.8463108          | 5.6501800000000001 | 0.262               |
| 7     | C    | GUMBEL_SOFTMAX | NONE      | 2.2927634          | 0.04886712982465819  | 1.8347742          | 6.924899999999999  | 0.42554             |
| 7     | C    | NONE           | NONE      | 2.4670753999999997 | 0.11414553813136985  | 1.8826764          | 7.0                | 0.36668             |
| 7     | C    | SOFT_GATING    | ELBO      | 2.3322602          | 0.0635044290518385   | 1.8453538000000003 | 3.7245199999999996 | 0.24582             |
| 7     | C    | SOFT_GATING    | K_PENALTY | 2.3184928          | 0.08161664132209792  | 1.8406718000000002 | 3.7196200000000004 | 0.24892000000000003 |
| 7     | C    | SOFT_GATING    | NONE      | 2.2988402          | 0.0716130432163024   | 1.8372742          | 5.6401200000000005 | 0.33214             |

bimodal k\_max=3

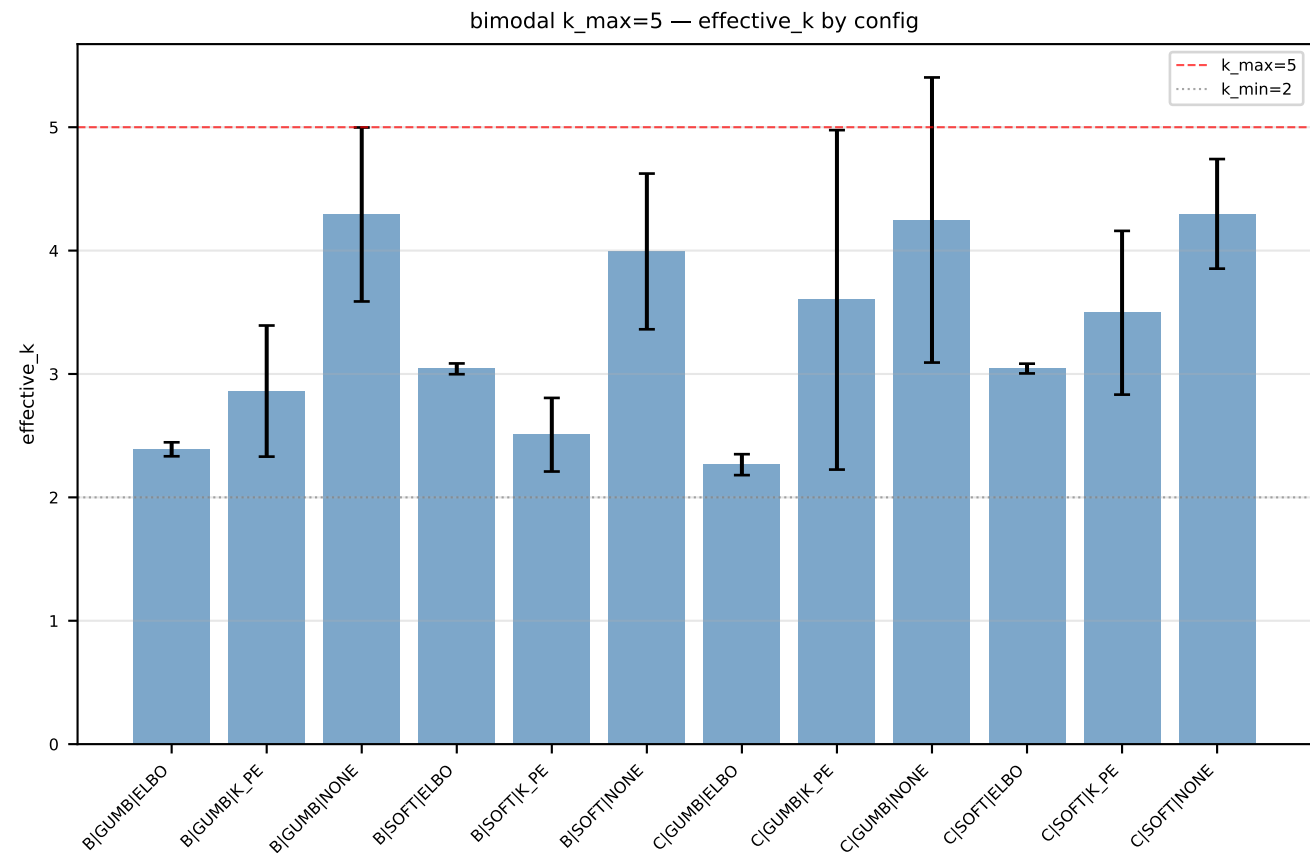
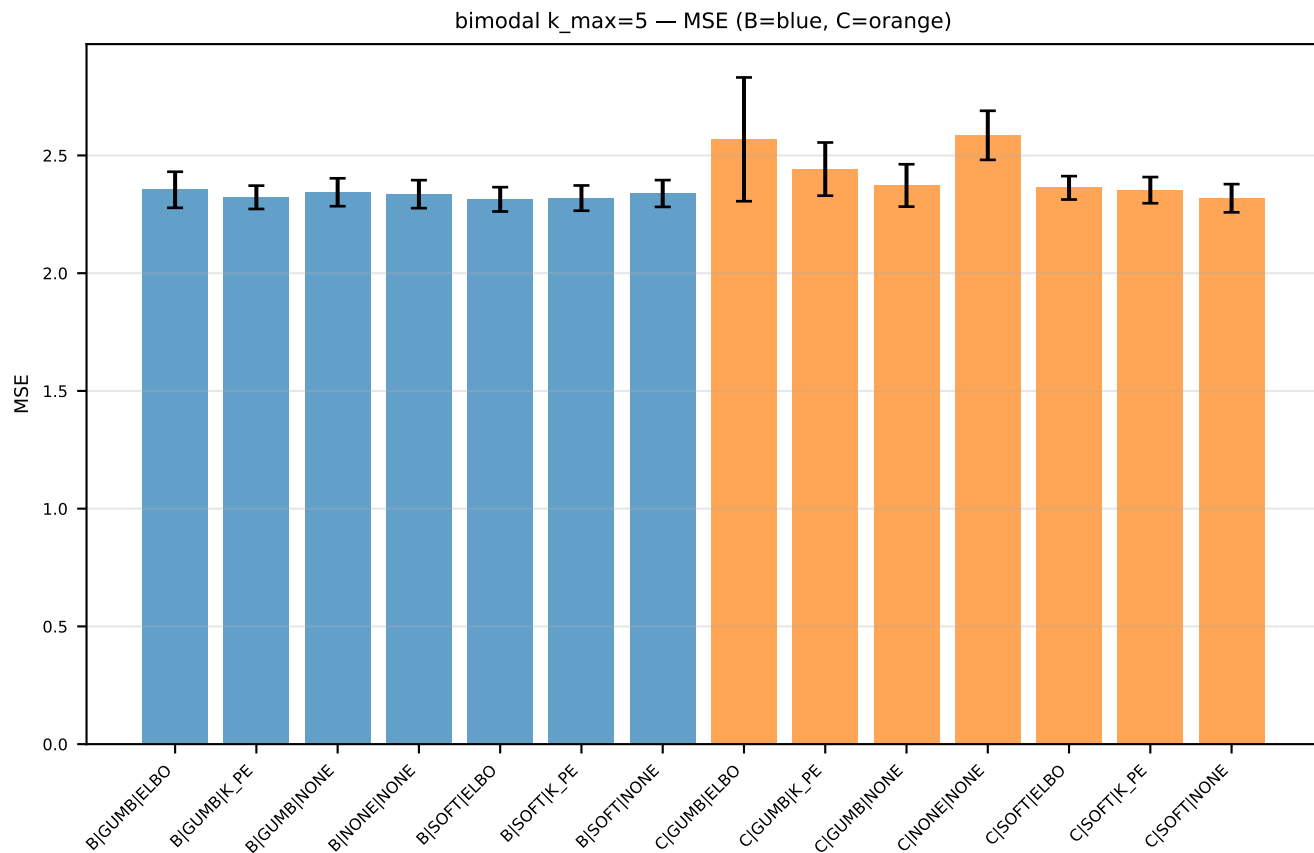
bimodal k\_max=3 — MSE (B=blue, C=orange)



bimodal k\_max=3 — effective\_k by config

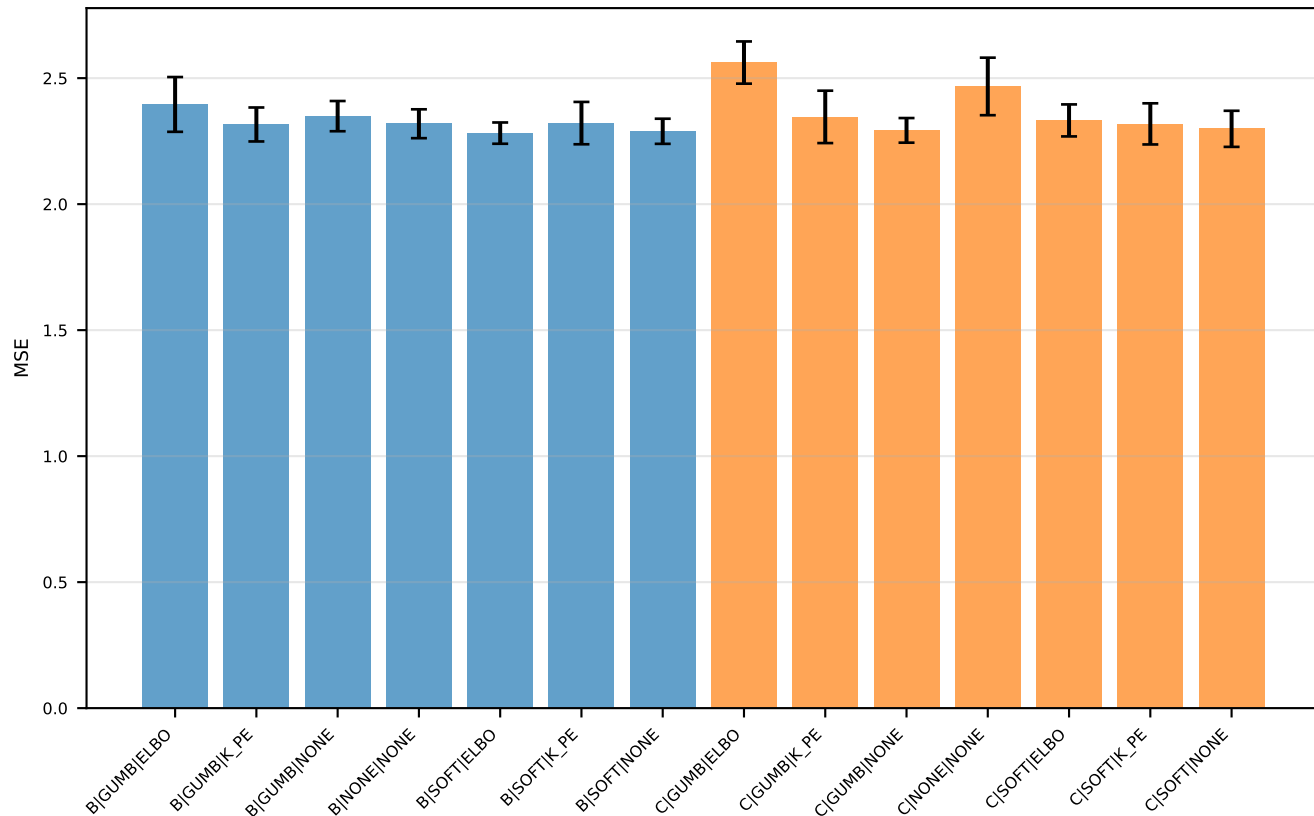


bimodal k\_max=5

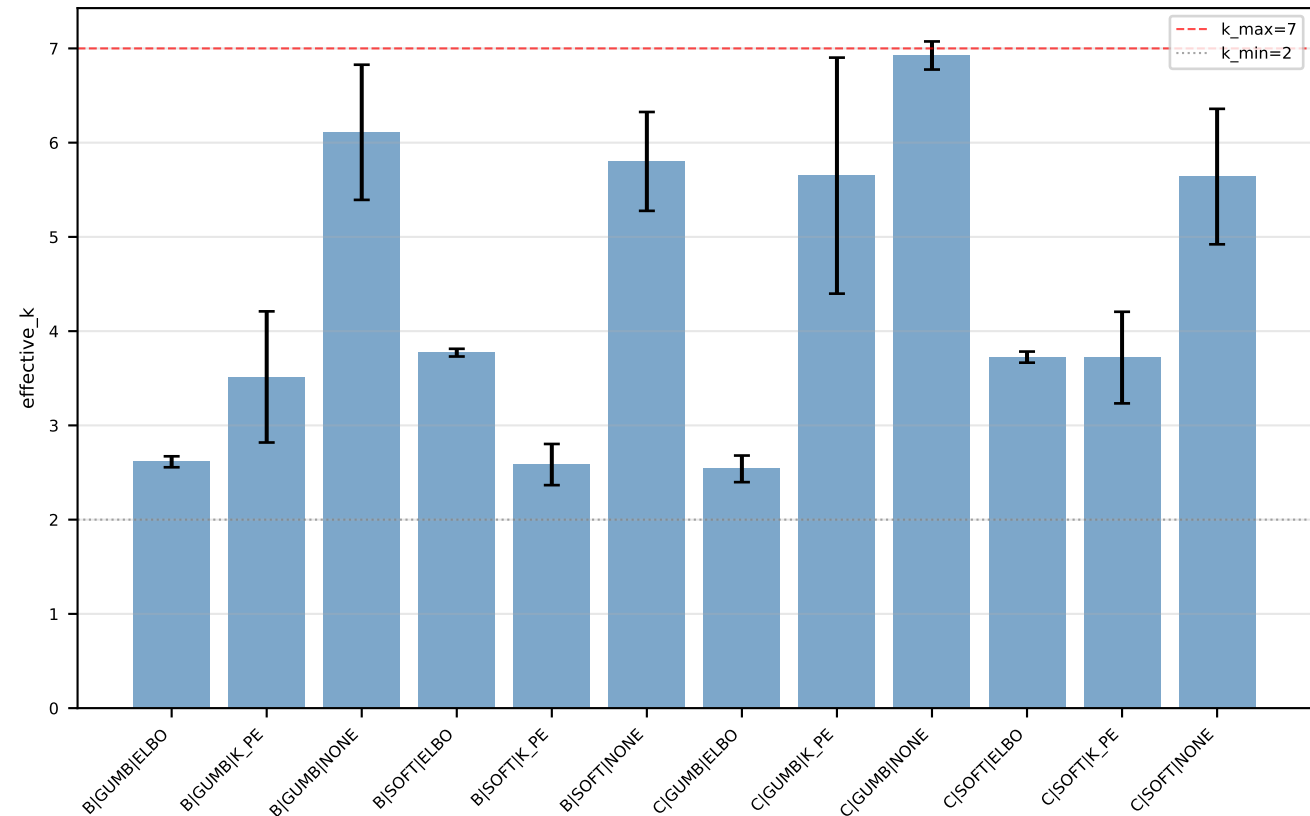


bimodal k\_max=7

bimodal k\_max=7 — MSE (B=blue, C=orange)



bimodal k\_max=7 — effective\_k by config

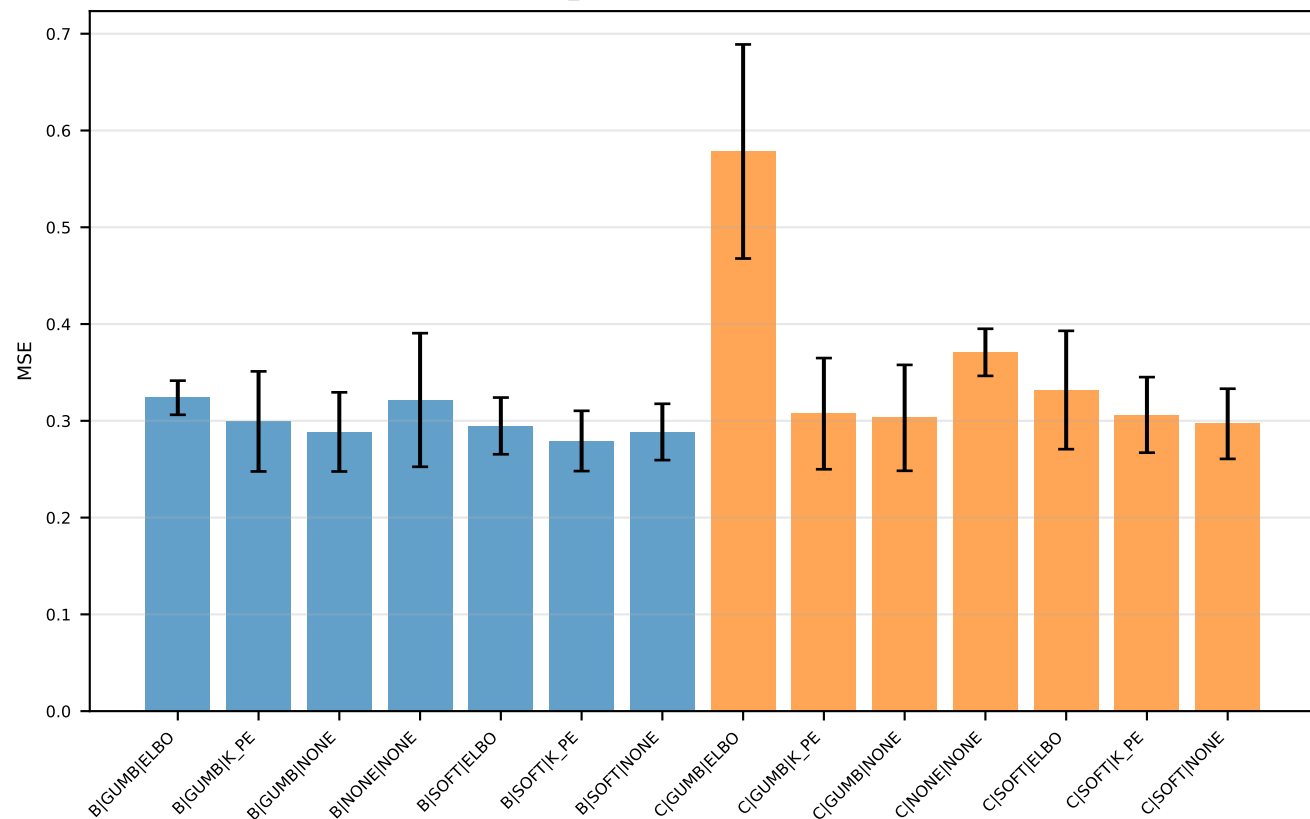




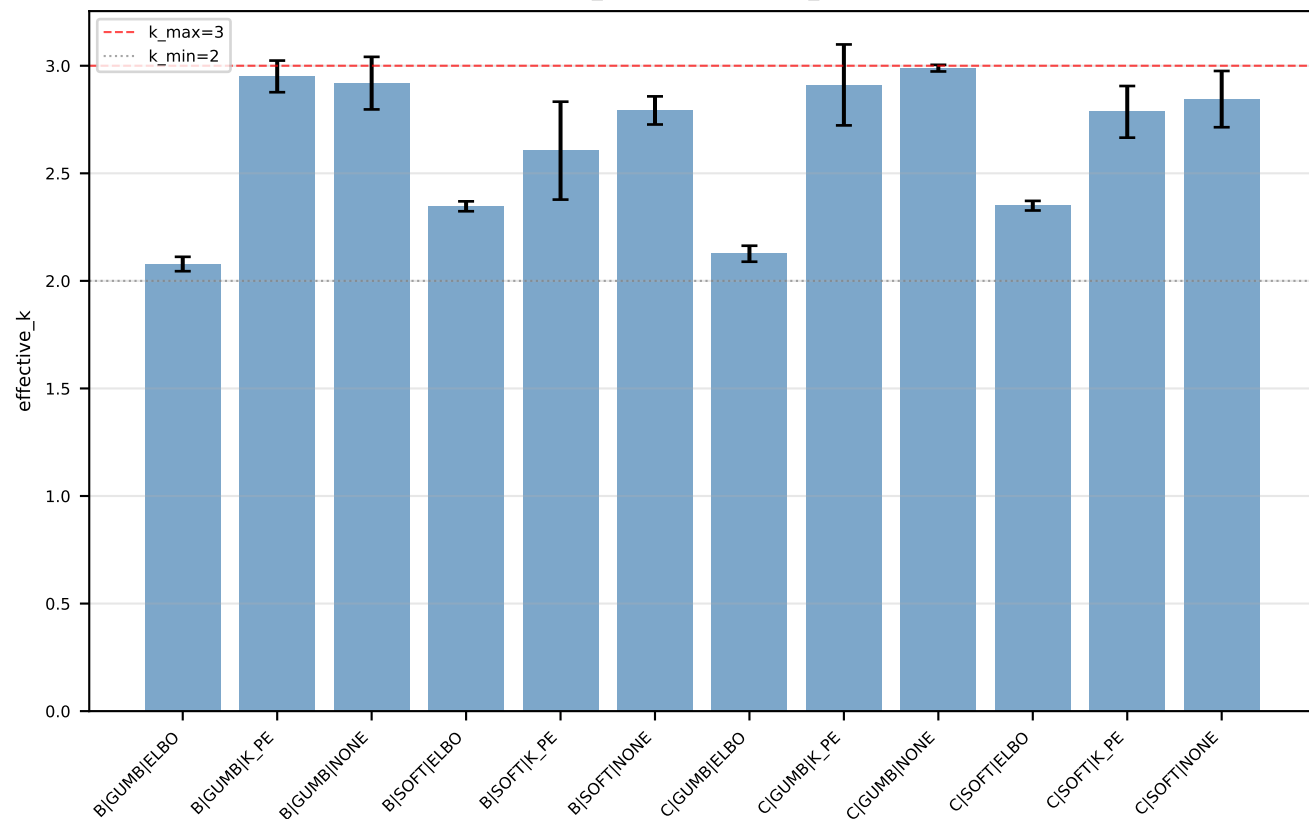
| k_max | cell | dynamic        | k_reg     | mse_mean                                           | mse_std              | nll_gaussian_mean   | effective_k_mean   | max_p_mid_mean      |
|-------|------|----------------|-----------|----------------------------------------------------|----------------------|---------------------|--------------------|---------------------|
| 3     | B    | GUMBEL_SOFTMAX | ELBO      | 0.32386360000000003                                | 0.017636688657454935 | 0.7513797999999999  | 2.0784             | 0.17786             |
| 3     | B    | GUMBEL_SOFTMAX | K_PENALTY | 0.2993812                                          | 0.05170685689248574  | 0.5322266           | 2.9505             | 0.16598             |
| 3     | B    | GUMBEL_SOFTMAX | NONE      | 0.2885436<br>piecewise — mean ± std across 5 seeds | 0.040886912934336335 | 0.5570094           | 2.91918            | 0.17284             |
| 3     | B    | NONE           | NONE      |                                                    | 0.06903228744797613  | 0.6012914           | 3.0                | 0.15608             |
| 3     | B    | SOFT_GATING    | ELBO      | 0.294735                                           | 0.02930793320075641  | 0.5437162000000001  | 2.34666            | 0.0722              |
| 3     | B    | SOFT_GATING    | K_PENALTY | 0.2791928                                          | 0.03113767310991622  | 0.47633580000000003 | 2.60538            | 0.14628000000000002 |
| 3     | B    | SOFT_GATING    | NONE      | 0.2884922                                          | 0.029095652109894357 | 0.47720219999999997 | 2.7922800000000003 | 0.16598             |
| 3     | C    | GUMBEL_SOFTMAX | ELBO      | 0.5783134                                          | 0.1106123049068231   | 1.087663            | 2.1262             | 0.1983              |
| 3     | C    | GUMBEL_SOFTMAX | K_PENALTY | 0.3073918                                          | 0.057455814542133166 | 0.5915914           | 2.91092            | 0.30718             |
| 3     | C    | GUMBEL_SOFTMAX | NONE      | 0.3030724                                          | 0.054723467354508906 | 0.5795044           | 2.98884            | 0.34696             |
| 3     | C    | NONE           | NONE      | 0.37076560000000003                                | 0.0243438886969376   | 0.8218954           | 3.0                | 0.33768000000000004 |
| 3     | C    | SOFT_GATING    | ELBO      | 0.3318296                                          | 0.06113972652294087  | 0.6731446           | 2.34952            | 0.3069              |
| 3     | C    | SOFT_GATING    | K_PENALTY | 0.3061136                                          | 0.03903855336459074  | 0.5257828           | 2.78576            | 0.30322             |
| 3     | C    | SOFT_GATING    | NONE      | 0.2969092                                          | 0.03626505433885353  | 0.5151703999999999  | 2.8449400000000002 | 0.35168             |
| 5     | B    | GUMBEL_SOFTMAX | ELBO      | 0.31158600000000003                                | 0.06441177879083917  | 0.5471672           | 2.35038            | 0.45410000000000006 |
| 5     | B    | GUMBEL_SOFTMAX | K_PENALTY | 0.2850348                                          | 0.02745625912610821  | 0.50238580000000001 | 3.8804600000000002 | 0.51394000000000001 |
| 5     | B    | GUMBEL_SOFTMAX | NONE      | 0.279553                                           | 0.02560100065427131  | 0.4953442           | 4.41744            | 0.5067999999999999  |
| 5     | B    | NONE           | NONE      | 0.29833940000000003                                | 0.03764287284201355  | 0.5597268           | 5.0                | 0.44845999999999997 |
| 5     | B    | SOFT_GATING    | ELBO      | 0.3011584                                          | 0.04988561446248807  | 0.5263658           | 3.0768999999999997 | 0.4723              |
| 5     | B    | SOFT_GATING    | K_PENALTY | 0.29473320000000003                                | 0.027791327994538144 | 0.4907876           | 3.7151400000000003 | 0.56278             |
| 5     | B    | SOFT_GATING    | NONE      | 0.3000064                                          | 0.04506615198571984  | 0.5117166           | 4.21586            | 0.43402             |
| 5     | C    | GUMBEL_SOFTMAX | ELBO      | 0.3972506                                          | 0.03515209199037803  | 0.8595273999999999  | 2.30314            | 0.308               |
| 5     | C    | GUMBEL_SOFTMAX | K_PENALTY | 0.36375159999999995                                | 0.11166291922254226  | 0.6470157999999999  | 4.4825800000000005 | 0.35338             |
| 5     | C    | GUMBEL_SOFTMAX | NONE      | 0.3705342                                          | 0.13574597195755017  | 0.633457            | 4.49938            | 0.37344             |
| 5     | C    | NONE           | NONE      | 0.3054224                                          | 0.0328592317667349   | 0.6261699999999999  | 5.0                | 0.45042             |
| 5     | C    | SOFT_GATING    | ELBO      | 0.31300540000000004                                | 0.06102345938899237  | 0.5556058           | 3.11928            | 0.30998000000000003 |
| 5     | C    | SOFT_GATING    | K_PENALTY | 0.3397204                                          | 0.08605091087431904  | 0.629645            | 3.80118            | 0.38312             |
| 5     | C    | SOFT_GATING    | NONE      | 0.3282816                                          | 0.11435081554715736  | 0.584013            | 4.1872             | 0.53276             |
| 7     | B    | GUMBEL_SOFTMAX | ELBO      | 0.3210868                                          | 0.04496217385425221  | 0.606513            | 2.63068            | 0.414               |
| 7     | B    | GUMBEL_SOFTMAX | K_PENALTY | 0.2936574                                          | 0.022126104860548782 | 0.5341718           | 5.94872            | 0.40352             |
| 7     | B    | GUMBEL_SOFTMAX | NONE      | 0.29397419999999996                                | 0.038796519414761935 | 0.5054486           | 6.3823             | 0.41646             |
| 7     | B    | NONE           | NONE      | 0.31921600000000006                                | 0.05357344716835012  | 0.5928114           | 7.0                | 0.14544             |
| 7     | B    | SOFT_GATING    | ELBO      | 0.30870319999999996                                | 0.04496708100043853  | 0.5135344           | 3.7495199999999995 | 0.48507999999999996 |
| 7     | B    | SOFT_GATING    | K_PENALTY | 0.2911674                                          | 0.03016631736390773  | 0.49086420000000003 | 4.39806            | 0.35316000000000003 |
| 7     | B    | SOFT_GATING    | NONE      | 0.28209819999999997                                | 0.039581441550302326 | 0.50081             | 5.22378            | 0.35302             |
| 7     | C    | GUMBEL_SOFTMAX | ELBO      | 0.3592932                                          | 0.02419734898082847  | 0.77847             | 2.58556            | 0.16934             |
| 7     | C    | GUMBEL_SOFTMAX | K_PENALTY | 0.366339                                           | 0.12873357902855023  | 0.621887            | 5.35262            | 0.2303              |
| 7     | C    | GUMBEL_SOFTMAX | NONE      | 0.31772039999999996                                | 0.03378213078389225  | 0.5930104           | 6.07372            | 0.3287              |
| 7     | C    | NONE           | NONE      | 0.2997064                                          | 0.030305297322415437 | 0.56223980000000001 | 7.0                | 0.31196             |
| 7     | C    | SOFT_GATING    | ELBO      | 0.3003018                                          | 0.05084827313390298  | 0.6956692           | 3.8091999999999997 | 0.32238             |
| 7     | C    | SOFT_GATING    | K_PENALTY | 0.31594540000000004                                | 0.05845380395919499  | 0.6974022           | 4.77742            | 0.21417999999999998 |
| 7     | C    | SOFT_GATING    | NONE      | 0.3086392                                          | 0.0559302289348077   | 0.6157368           | 5.13048            | 0.22864             |

piecewise k\_max=3

piecewise k\_max=3 — MSE (B=blue, C=orange)

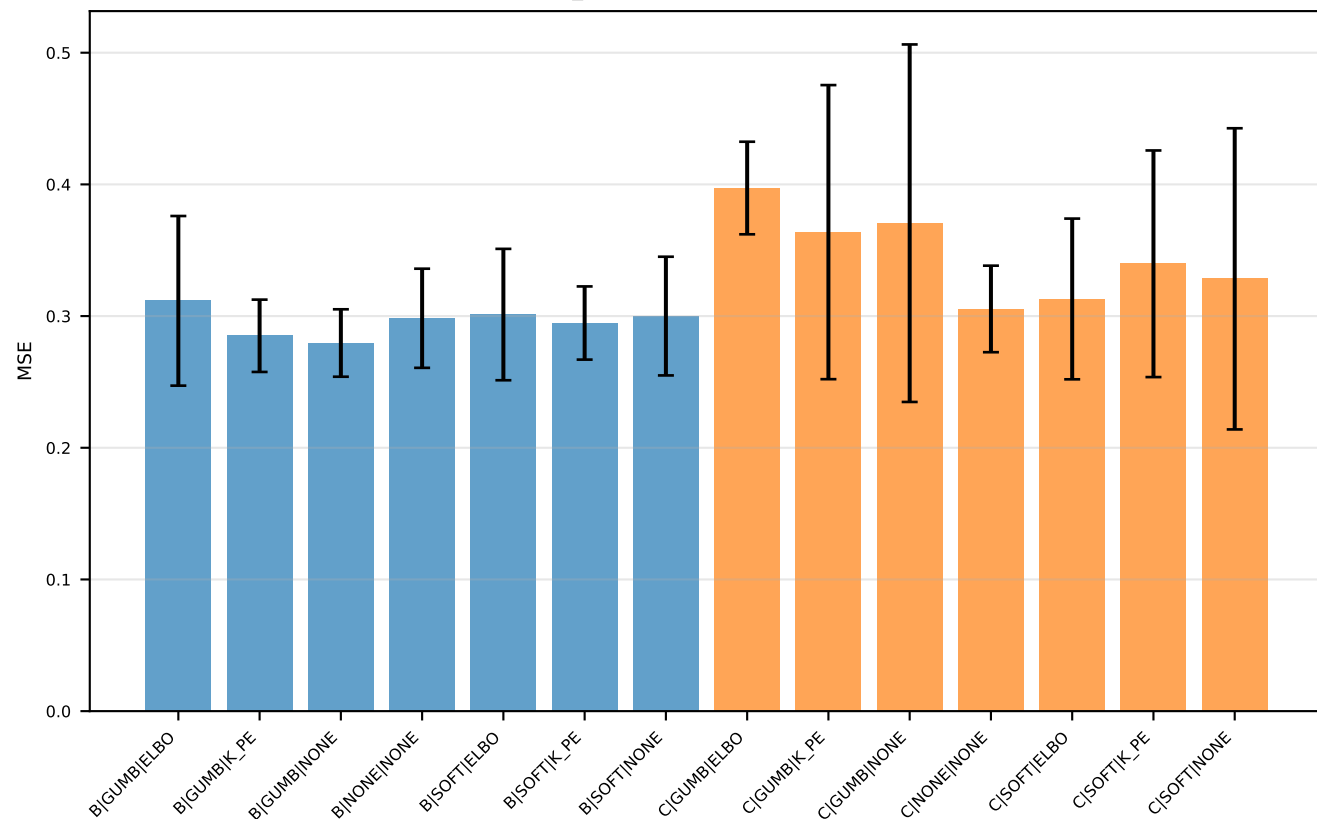


piecewise k\_max=3 — effective\_k by config

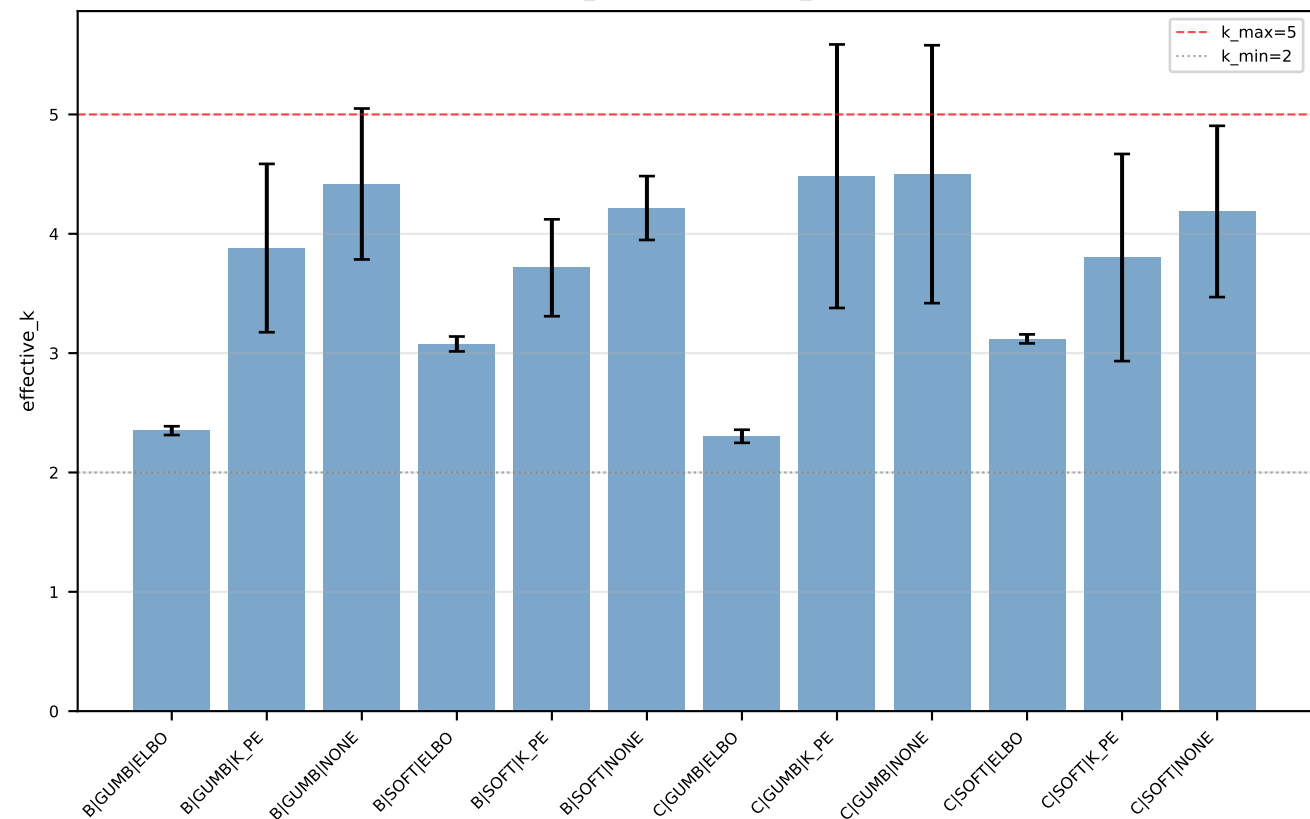


piecewise k\_max=5

piecewise k\_max=5 — MSE (B=blue, C=orange)

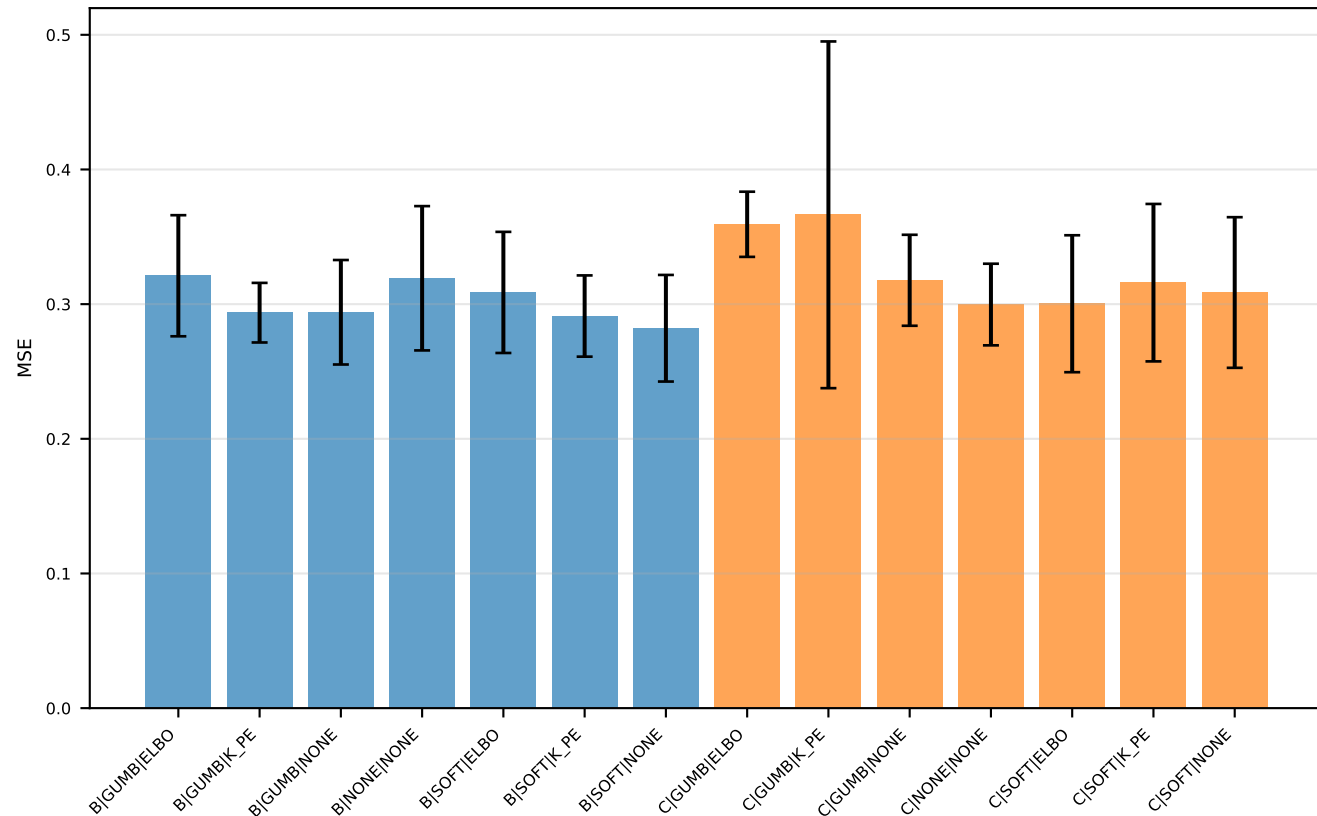


piecewise k\_max=5 — effective\_k by config

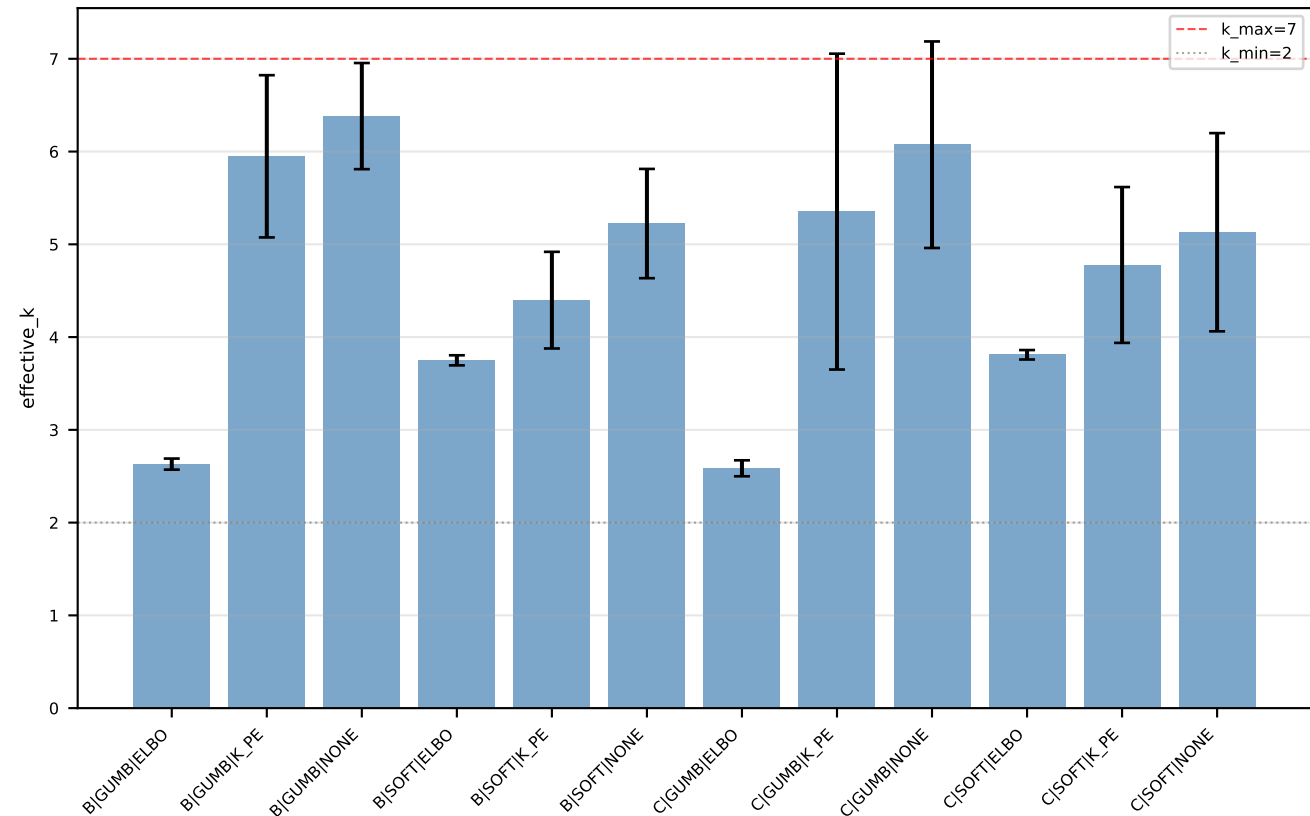


piecewise k\_max=7

piecewise k\_max=7 — MSE (B=blue, C=orange)



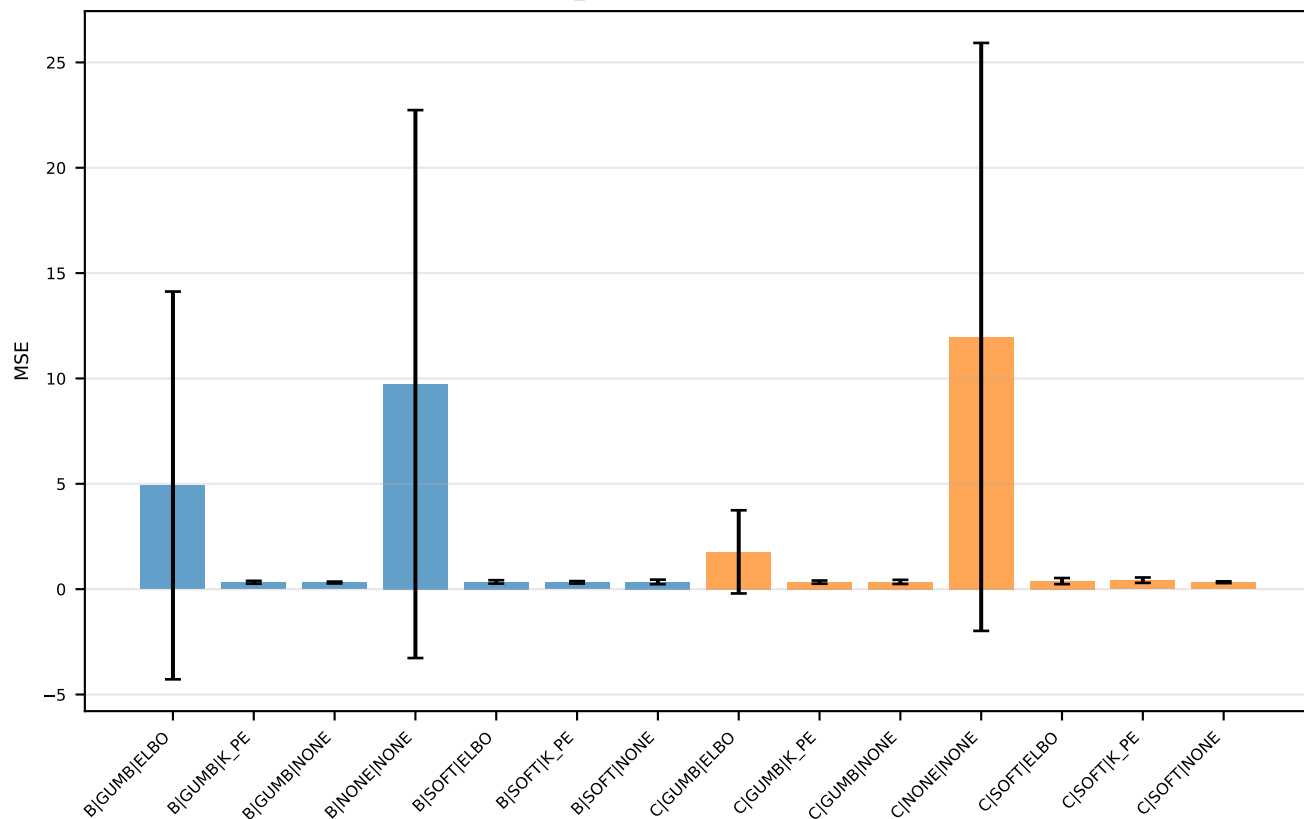
piecewise k\_max=7 — effective\_k by config



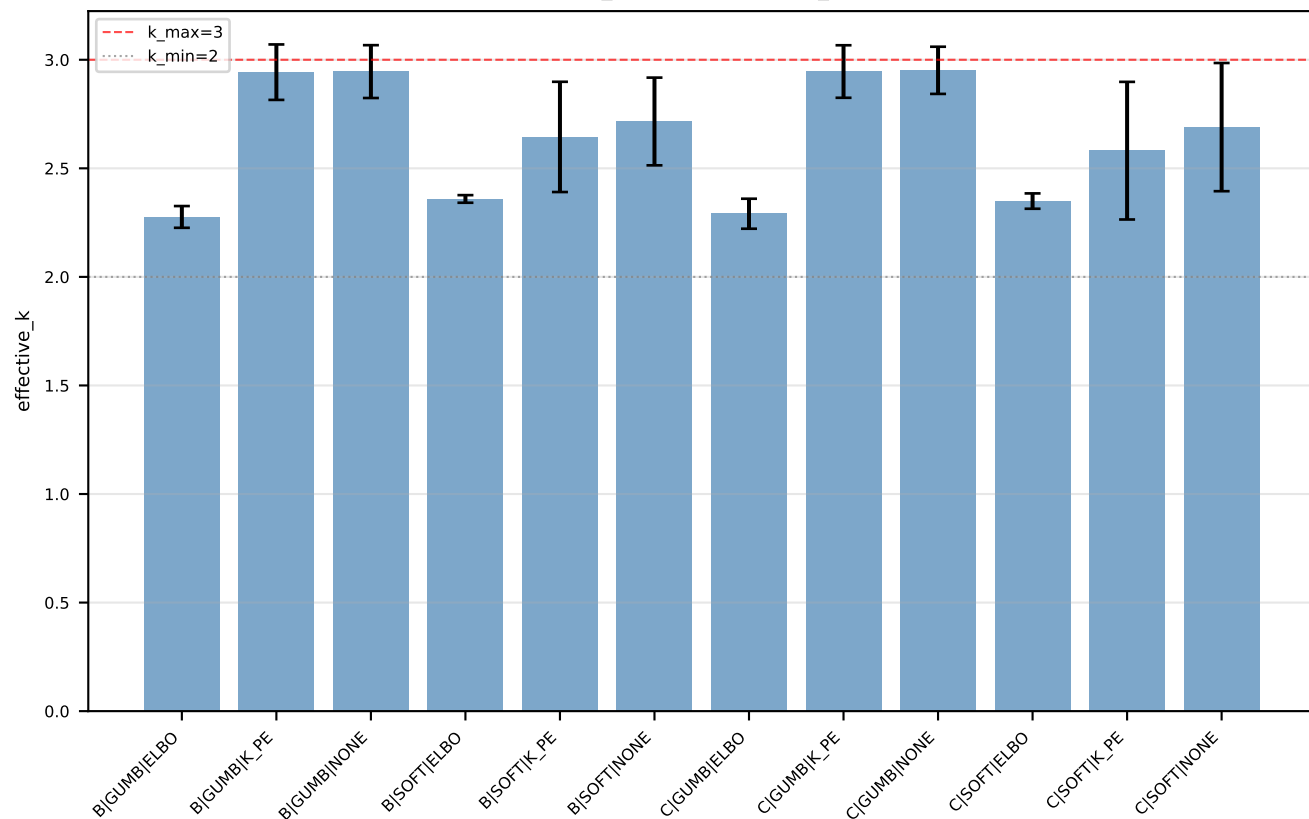
| k_max | cell | dynamic        | k_reg     | mse_mean            | mse_std              | nll_gaussian_mean   | effective_k_mean   | max_p_mid_mean      |
|-------|------|----------------|-----------|---------------------|----------------------|---------------------|--------------------|---------------------|
| 3     | B    | GUMBEL_SOFTMAX | ELBO      | 4.9228152000000005  | 9.202026299194419    | 1.3147812           | 2.27616            | 0.33942             |
| 3     | B    | GUMBEL_SOFTMAX | K_PENALTY | 0.3271372           | 0.06858735435705333  | 0.8475509999999999  | 2.94286            | 0.72314             |
| 3     | B    | GUMBEL_SOFTMAX | NONE      | 0.3153362           | 0.043862187983501236 | 0.8504764           | 2.94556            | 0.72324             |
| 3     | B    | NONE           | NONE      | 9.7342172           | 13.003137980850802   | 1.5056812           | 3.0                | 0.4615399999999995  |
| 3     | B    | SOFT_GATING    | ELBO      | 0.344004            | 0.0814755263284626   | 0.8909174           | 2.35902            | 0.4358000000000001  |
| 3     | B    | SOFT_GATING    | K_PENALTY | 0.32387299999999997 | 0.05878229422623788  | 0.8701753999999999  | 2.64472            | 0.6351600000000001  |
| 3     | B    | SOFT_GATING    | NONE      | 0.34440980000000004 | 0.10782402447367655  | 0.8678096           | 2.7158             | 0.61218             |
| 3     | C    | GUMBEL_SOFTMAX | ELBO      | 1.769714            | 1.9742066778716454   | 1.2135158000000001  | 2.2909             | 0.17934             |
| 3     | C    | GUMBEL_SOFTMAX | K_PENALTY | 0.33355480000000004 | 0.07627666123854664  | 0.8504455999999999  | 2.94588            | 0.27658             |
| 3     | C    | GUMBEL_SOFTMAX | NONE      | 0.3440004           | 0.09863042353554001  | 0.8604796           | 2.9514             | 0.28838             |
| 3     | C    | NONE           | NONE      | 11.972039           | 13.953123610119619   | 1.6706654           | 3.0                | 0.3657              |
| 3     | C    | SOFT_GATING    | ELBO      | 0.3832254           | 0.1448880780267997   | 0.9300264           | 2.34918            | 0.19524             |
| 3     | C    | SOFT_GATING    | K_PENALTY | 0.4262646           | 0.12818038851282987  | 0.9181782           | 2.58128            | 0.3235              |
| 3     | C    | SOFT_GATING    | NONE      | 0.32886540000000003 | 0.04277998337774339  | 0.8568232           | 2.6899             | 0.31646             |
| 5     | B    | GUMBEL_SOFTMAX | ELBO      | 7.3044101999999995  | 13.699328880638522   | 1.3892974           | 2.41238            | 0.7071799999999999  |
| 5     | B    | GUMBEL_SOFTMAX | K_PENALTY | 0.6118554           | 0.3816495944256983   | 1.0322324           | 4.8803600000000005 | 0.7021              |
| 5     | B    | GUMBEL_SOFTMAX | NONE      | 0.5045628           | 0.24412773852370814  | 1.0308432           | 4.94692            | 0.72224             |
| 5     | B    | NONE           | NONE      | 5.9089304000000001  | 4.912415070525016    | 1.2481740000000001  | 5.0                | 0.65538             |
| 5     | B    | SOFT_GATING    | ELBO      | 0.3259046           | 0.06649200863111294  | 0.8569668           | 3.15756            | 0.68994             |
| 5     | B    | SOFT_GATING    | K_PENALTY | 0.34006020000000003 | 0.06910713342846742  | 0.8736054           | 4.0239400000000005 | 0.73636             |
| 5     | B    | SOFT_GATING    | NONE      | 0.327779            | 0.04190344349215229  | 0.8691407999999999  | 4.645239999999999  | 0.72802             |
| 5     | C    | GUMBEL_SOFTMAX | ELBO      | 5.6660242000000001  | 3.2990035016982326   | 1.6379106           | 2.4614000000000003 | 0.34074             |
| 5     | C    | GUMBEL_SOFTMAX | K_PENALTY | 0.31343160000000003 | 0.037264854532119154 | 0.8430440000000001  | 4.9971000000000005 | 0.38828             |
| 5     | C    | GUMBEL_SOFTMAX | NONE      | 0.3384704           | 0.038389703968642414 | 0.8681787999999999  | 4.99666            | 0.4516799999999997  |
| 5     | C    | NONE           | NONE      | 2.4200104           | 0.6820573480194315   | 1.2973203999999998  | 5.0                | 0.46902             |
| 5     | C    | SOFT_GATING    | ELBO      | 0.96743260000000001 | 0.9785754235912529   | 0.9723186           | 3.12854            | 0.31120000000000003 |
| 5     | C    | SOFT_GATING    | K_PENALTY | 0.3140564           | 0.04349519223776348  | 0.8574702000000001  | 4.9573599999999995 | 0.542               |
| 5     | C    | SOFT_GATING    | NONE      | 0.3185478           | 0.04111041678446359  | 0.8508566           | 4.97874            | 0.46946000000000004 |
| 7     | B    | GUMBEL_SOFTMAX | ELBO      | 0.82885340000000001 | 0.6497909096088526   | 1.1363846           | 2.80092            | 0.32874             |
| 7     | B    | GUMBEL_SOFTMAX | K_PENALTY | 0.4487622           | 0.11167004749797503  | 0.999433            | 6.4815200000000001 | 0.4889              |
| 7     | B    | GUMBEL_SOFTMAX | NONE      | 0.4440836           | 0.1261883350524128   | 0.99712740000000001 | 6.6503             | 0.48672000000000004 |
| 7     | B    | NONE           | NONE      | 9.708612            | 11.863963656340385   | 1.3784208           | 7.0                | 0.5012399999999999  |
| 7     | B    | SOFT_GATING    | ELBO      | 0.3585434           | 0.08008926146681589  | 0.90916460000000001 | 3.84234            | 0.69944             |
| 7     | B    | SOFT_GATING    | K_PENALTY | 0.32581899999999997 | 0.02101283859929448  | 0.864422            | 4.918559999999999  | 0.5174              |
| 7     | B    | SOFT_GATING    | NONE      | 0.3131858           | 0.02582355004061216  | 0.8387283999999999  | 5.73398            | 0.39646             |
| 7     | C    | GUMBEL_SOFTMAX | ELBO      | 2.7412994           | 1.7621975721018912   | 1.3667696           | 2.7619             | 0.15638000000000002 |
| 7     | C    | GUMBEL_SOFTMAX | K_PENALTY | 0.5242794           | 0.3998889216648793   | 0.89952020000000001 | 6.583119999999999  | 0.38358             |
| 7     | C    | GUMBEL_SOFTMAX | NONE      | 0.3711664           | 0.1262416933417799   | 0.8730502           | 6.98612            | 0.33718             |
| 7     | C    | NONE           | NONE      | 1.0886206           | 0.3994652248497984   | 1.168259            | 7.0                | 0.32924000000000003 |
| 7     | C    | SOFT_GATING    | ELBO      | 1.09630740000000002 | 1.6501945672076976   | 0.9792772           | 3.83662            | 0.21502             |
| 7     | C    | SOFT_GATING    | K_PENALTY | 0.31013419999999997 | 0.023714616657243268 | 0.8339836           | 5.9427             | 0.36049999999999993 |
| 7     | C    | SOFT_GATING    | NONE      | 0.3166002           | 0.03509857481009733  | 0.8537279999999999  | 6.3504000000000005 | 0.25482             |

# exponential k\_max=3

exponential k\_max=3 — MSE (B=blue, C=orange)

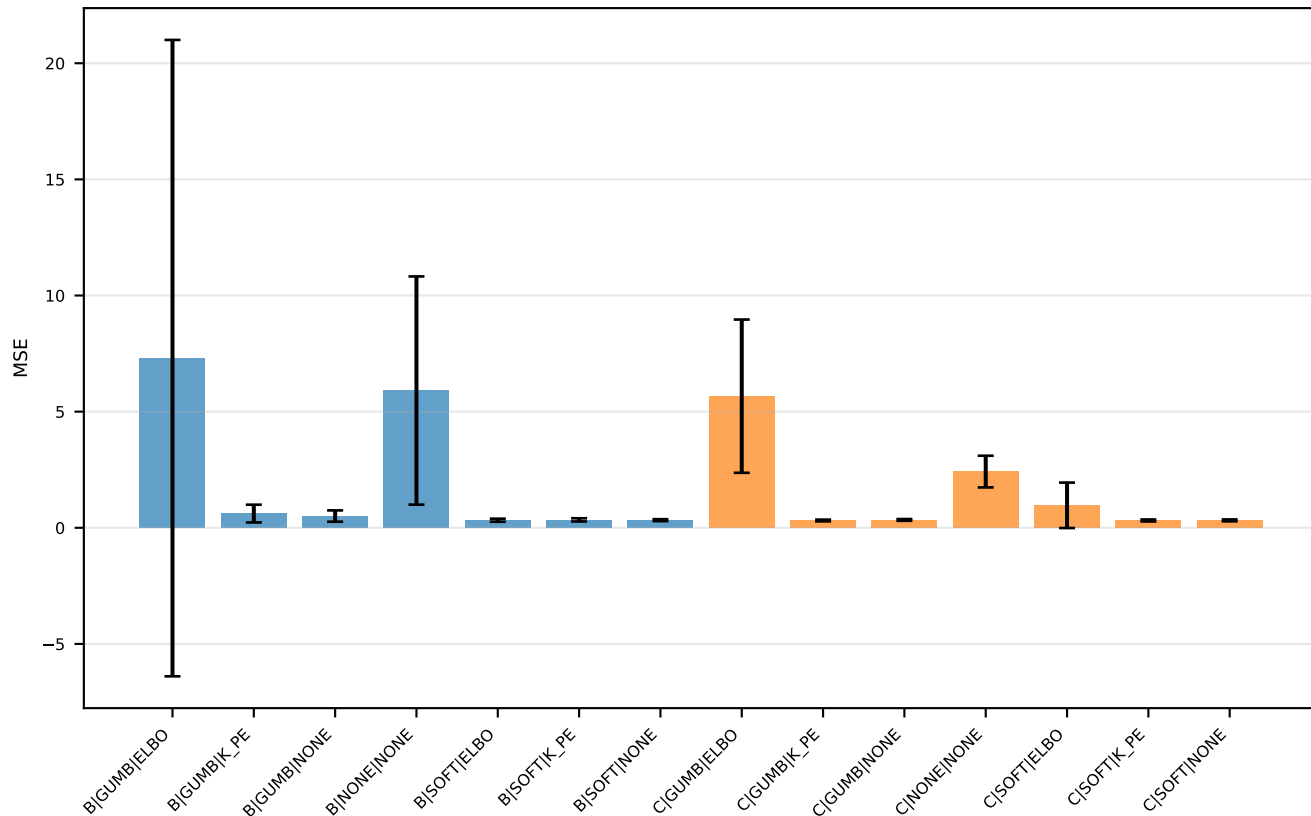


exponential k\_max=3 — effective\_k by config

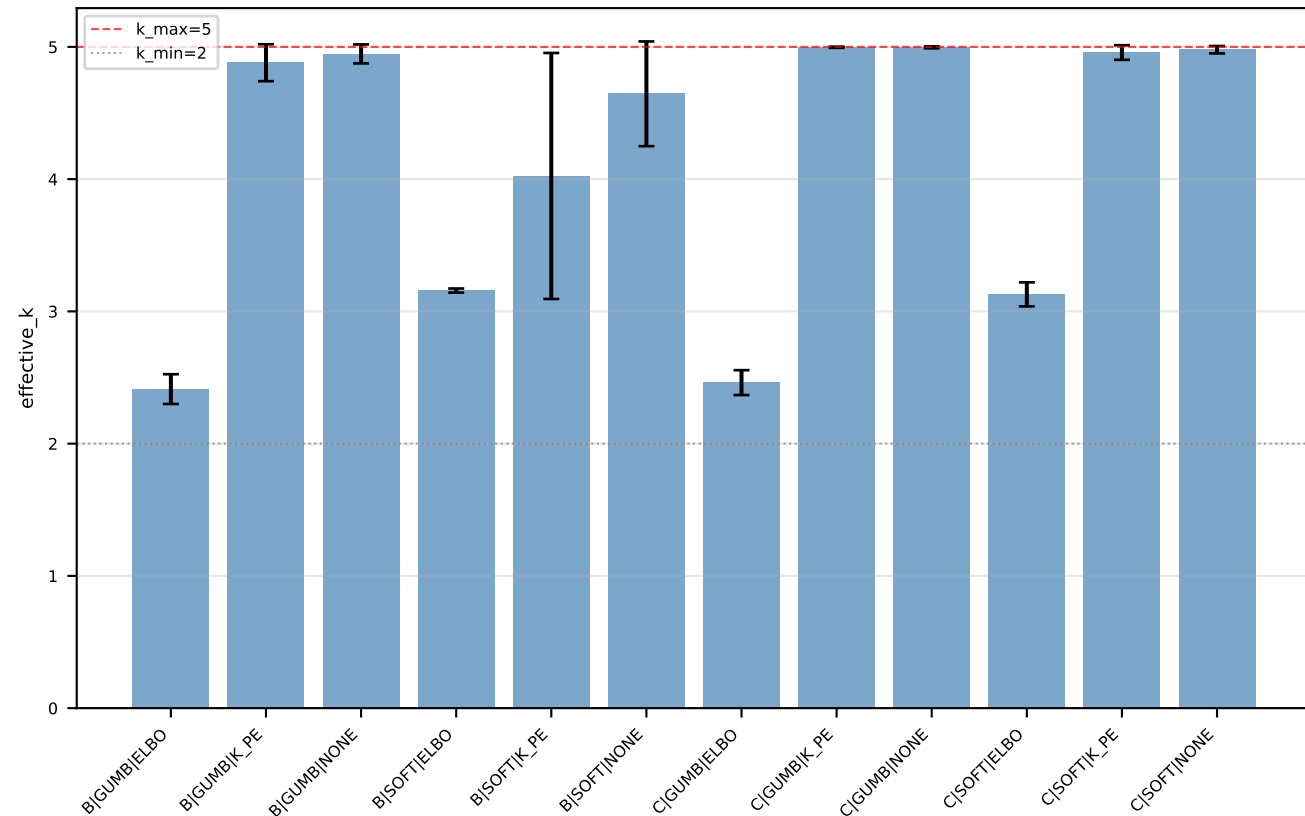


# exponential k\_max=5

exponential k\_max=5 — MSE (B=blue, C=orange)

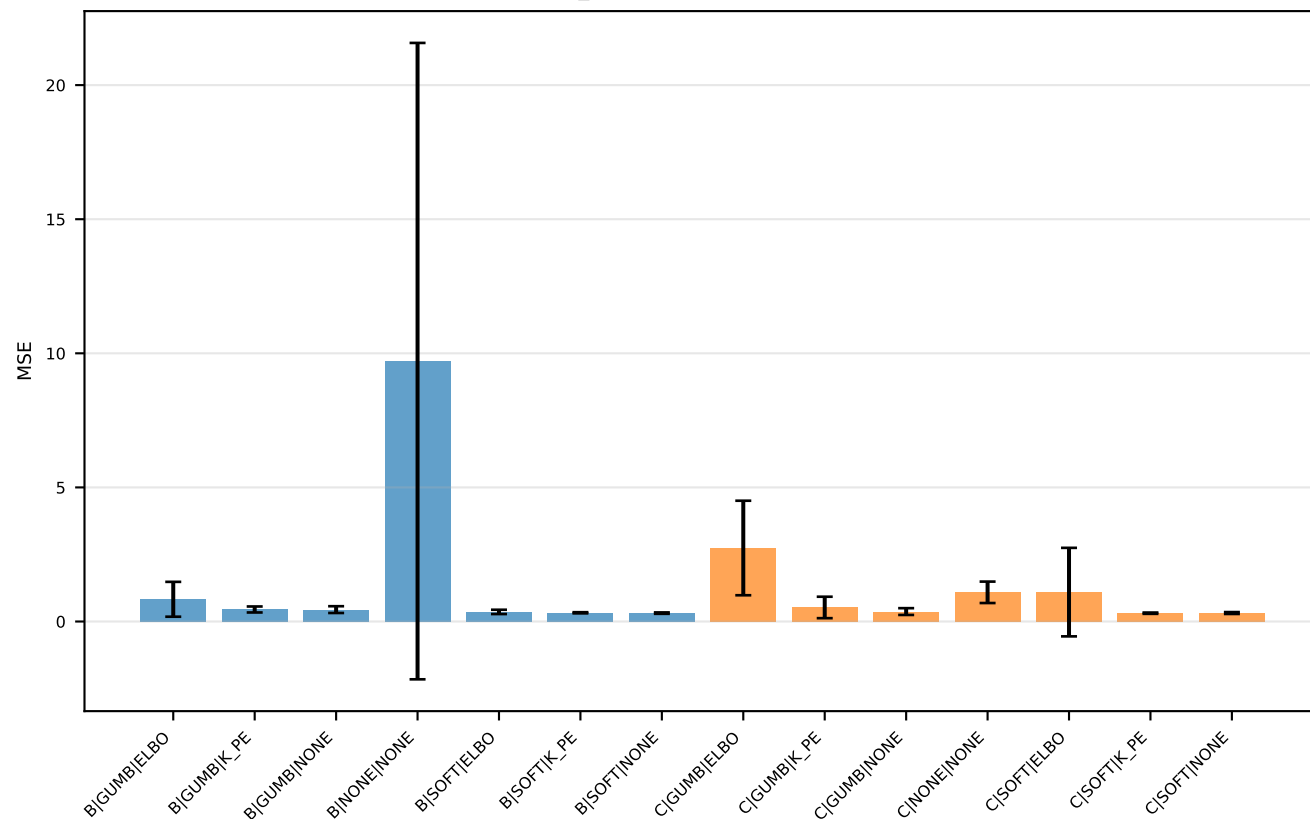


exponential k\_max=5 — effective\_k by config



# exponential k\_max=7

exponential k\_max=7 — MSE (B=blue, C=orange)



exponential k\_max=7 — effective\_k by config

